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Autophagosome development and chloroplast segmentation occur synchronously for piecemeal degradation of chloroplasts

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Abstract

Plants distribute many nutrients to chloroplasts during leaf development and maturation. When leaves senesce or experience sugar starvation, the autophagy machinery degrades chloroplast proteins to facilitate efficient nutrient reuse. Here, we report on the intracellular dynamics of an autophagy pathway responsible for piecemeal degradation of chloroplast components. Through live-cell monitoring of chloroplast morphology, we observed the formation of chloroplast budding structures in sugar-starved leaves. These buds were then released and incorporated into the vacuolar lumen as an autophagic cargo termed a Rubisco-containing body. The budding structures did not accumulate in mutants of core autophagy machinery, suggesting that autophagosome creation is required for forming chloroplast buds. Simultaneous tracking of chloroplast morphology and autophagosome development revealed that the isolation membranes of autophagosomes interact closely with part of the chloroplast surface before forming chloroplast buds. Chloroplasts then protrude at the site associated with the isolation membranes, which divide synchronously with autophagosome maturation. This autophagy-related division does not require DYNAMIN-RELATED PROTEIN 5B, which constitutes the division ring for chloroplast proliferation in growing leaves. An unidentified division machinery may thus fragment chloroplasts for degradation in coordination with the development of the chloroplast-associated isolation membrane.

eLife assessment

This manuscript investigates how chloroplasts are broken down during light-limiting conditions as plants reorganize their energy-producing organelles during carbon limitation. The authors provide **compelling** live-cell imaging data of plastids and **solid** quantification of events, documenting that buds form on the surface of chloroplasts and pinch away, then associate with the vacuole via a mechanism that depends on autophagy machinery, but not plastid division machinery. This manuscript provides **valuable** groundwork for other scientists studying the regulation and breakdown of energy-producing organelles, including chloroplasts and mitochondria.

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Introduction

Organelle morphology changes dynamically in response to fluctuations in cell functions, developmental stages, and environmental cues. Plastids are plant-specific organelles that differentiate into chloroplasts in green tissues to perform photosynthesis. Plastids further serve as hubs for metabolic pathways such as the biosynthesis of amino acids and plant hormones. Dynamic changes in plastid morphology also act as switches initiating biological programs (Osteryoung and Pyke, 2014 [↗](#)). For instance, chloroplasts in leaf pavement cells form stroma-filled tubular extensions termed stromules to activate programmed cell death as an immune response upon perception of an invading pathogen, in a process regulated by the plant hormone salicylic acid (SA) (Caplan et al., 2015 [↗](#)). Stromules are thin, tubular structures of plastid stroma less than 1 μm in diameter (Hanson and Sattarzadeh, 2011 [↗](#)).

Leaf aging is closely associated with morphological and functional changes of plastids. At the early stage of leaf development, the undifferentiated form of plastids, termed proplastids, are converted to green chloroplasts as their population expands through active division. In *Arabidopsis* (*Arabidopsis thaliana*), the coordination of cytoplasmic ring structures comprising DYNAMIN-RELATED PROTEIN 5B (DRP5B), also called ACCUMULATION AND REPLICATION OF CHLOROPLASTS 5 (ARC5), and intrachloroplastic ring structures consisting of FILAMENTING TEMPERATURE-SENSITIVE Z (FTSZ) proteins mediates chloroplast division (Chen et al., 2018 [↗](#)). In mature leaves, spherical chloroplasts occupy almost all of the cytoplasm and manifest high photosynthetic activity. When older leaves later senesce, the size and number of chloroplasts gradually decrease (Martinoia et al., 1983 [↗](#); Mae et al., 1984 [↗](#)). Younger leaves closer to the shoot apex sometimes shade older leaves beneath them, thus perturbing photosynthesis by decreasing the amount of light reaching those leaves and causing local sugar starvation. In such sugar-starved leaves, senescence is accelerated to facilitate the relocation of nutrients into newly developing tissues (Weaver and Amasino, 2001 [↗](#); Ono et al., 2013 [↗](#); Law et al., 2018 [↗](#)). Autophagy is an intracellular degradation machinery that contributes to the decline in chloroplast volume and numbers during leaf senescence (Wada et al., 2009 [↗](#)).

Autophagy is a ubiquitous mechanism that transports cytoplasmic components to the vacuolar/lysosomal compartment for degradation in eukaryotic cells. The major autophagy pathway known as macroautophagy, or simply autophagy, begins with the assembly of a membrane termed the isolation membrane (also known as a phagophore) that forms a double-membrane structure termed the autophagosome and sequesters a portion of the cytoplasm. Core AUTOPHAGY (ATG) proteins (ATG1–10, 12–14, 16, and 18) are highly conserved among plants,

yeasts, and mammals (Yoshimoto and Ohsumi, 2018 [↗](#); Nakamura et al., 2021a [↗](#)) and are required for the initiation and elongation of the isolation membrane (Nakatogawa, 2020 [↗](#)). The outer membrane of autophagosomes fuses with the vacuolar/lysosomal membrane, resulting in the formation of autophagic bodies whose cargos are digested by vacuolar/lysosomal hydrolases. Our previous studies identified two types of autophagy for chloroplast degradation in mature leaves (Ishida et al., 2014 [↗](#); Nakamura and Izumi, 2018 [↗](#)). Senescence and sugar starvation preferentially activate piecemeal degradation of chloroplast components involving a specific type of autophagosomal cargo termed a Rubisco-containing body (RCB), leading to smaller chloroplasts (Ishida et al., 2008 [↗](#); Wada et al., 2009 [↗](#); Izumi et al., 2015 [↗](#)). The second type of autophagy, termed chlorophagy, removes entire unnecessary chloroplasts (Izumi et al., 2017 [↗](#); Nakamura et al., 2018 [↗](#)), thereby modulating the number of chloroplasts in a cell (Wada et al., 2009 [↗](#); Izumi et al., 2017 [↗](#)). Chlorophagy is likely another type of autophagy, termed microautophagy, during which the vacuolar membrane directly participates in sequestering the degradation target in the absence of encapsulation by autophagosomes (Nakamura et al., 2017; Lee et al., 2023 [↗](#)).

Mitochondria are another type of energy-producing organelle derived from endosymbiosis. Dynamin-related proteins (DRPs) also mediate mitochondrial division (Giacomello et al., 2020 [↗](#)); the involvement of DRP-mediated organelle division in mitochondrion-targeted autophagy—termed mitophagy—in mammals and budding yeast (*Saccharomyces cerevisiae*) has been under debate (Chen et al., 2022 [↗](#)). Drp1 and Dnm1 (Dynamin-related 1) participate in mitochondrial fission in mammals and budding yeast, respectively. Previous studies have suggested that segmented mitochondria resulting from Drp1/Dnm1-mediated fission become a target for degradation by autophagosomes (Twig et al., 2008 [↗](#); Rambold et al., 2011 [↗](#); Abeliovich et al., 2013 [↗](#)). Notably, Drp1/Dnm1-independent mitophagy has also been observed (Yamashita et al., 2016 [↗](#)), whereby mitochondrial division occurs concomitantly with the development of the mitochondrion-associated isolation membrane, forming an autophagosome that specifically contains mitochondrial components termed the mitophagosome. A recent study identified the mitochondrial intermembrane-space protein mitofissin (also called Atg44), required for mitochondrial fission during yeast mitophagy (Fukuda et al., 2023 [↗](#)).

Plant autophagic bodies and RCBs are typically around 1 μm in diameter (Chiba et al., 2003 [↗](#); Yoshimoto et al., 2004 [↗](#)), which is smaller than chloroplasts. Thus, piecemeal-type chloroplast autophagy via RCBs must start with division of the chloroplast segments that are to be degraded. However, how a fragment of a chloroplast can be transported to the vacuole remains poorly understood. Here, to characterize the underlying intracellular dynamics, we performed high-resolution imaging analyses in living cells of *Arabidopsis* leaves. We used time-lapse microscopy to observe the segmentation of chloroplast budding structures and their transport to the vacuolar lumen. Use of multiple organelle markers revealed that the development of the chloroplast-associated isolation membrane and the formation of RCBs occur simultaneously. These morphological changes did not correspond to the emergence of stromules or to DRP5b-mediated chloroplast division. Therefore, a previously undescribed chloroplast division machinery may be in play during chloroplast autophagy.

Results

Chloroplast budding structures containing stroma and envelope are released in response to sugar starvation

To observe the intracellular dynamics of the piecemeal-type of chloroplast autophagy, we subjected the mature leaves of *Arabidopsis* plants to incubation in darkness, as this treatment induces sugar starvation and accelerates leaf senescence, thus activating the RCB-mediated chloroplast autophagy (Izumi et al., 2010 [↗](#)). Exogenous application of concanamycin A (conCA), an inhibitor of vacuolar H^+ -ATPase, allows the vacuolar accumulation of autophagic cargos, including

RCBs, facilitating their visualization (Yoshimoto et al., 2004; Ishida et al., 2008). Following the excision and incubation in darkness of leaves harboring chloroplast stroma-targeted fluorescent protein markers such as RUBISCO SMALL SUBUNIT (RBCS) fused to monomeric red fluorescent protein (mRFP) in the presence of concA, we observed the accumulation of many small puncta containing the RBCS-mRFP signal in the vacuole, i.e., RCBs (**Figure 1-figure supplement 1A** and **1C**). Although co-incubation of dark-treated leaves with concA and mineral nutrient-rich Murashige and Skoog (MS) salts did not block the RCB accumulation, the addition of sucrose did, by rescuing the sugar-starved leaf (**Figure 1-figure supplement 1A** and **1C**). We detected no RCBs when we subjected the leaves of a mutant for the core *ATG* gene *ATG7* to dark incubation in sucrose-free solution (**Figure 1-figure supplement 1B** and **1C**). These observations indicate that RCBs are a type of autophagic cargo involved in the degradation of chloroplast fragments in response to sugar starvation. Therefore, in this study, we monitored chloroplast morphology in sugar-starved leaves resulting from their incubation in darkness in a sugar-free solution.

We wished to monitor the changes in the three-dimensional (3D) morphology of chloroplasts during autophagic degradation of chloroplast fragments. We thus used a two-photon excitation microscope equipped with a confocal spinning-disk unit (Otomo et al., 2015), which facilitates the monitoring of changes in the 3D cellular structures of living plant cells (Sasaki et al., 2019; Nakamura et al., 2021b). Using dark-incubated leaves from transgenic plants harboring chloroplast-targeted *GFP* (*CT-GFP*) or *RBCS-mRFP* transgenes, we determined that mesophyll chloroplasts form a type of budding structure containing stromal components (**Figure 1A** and **1B**, arrowheads). We observed these structures budding off from their associated chloroplasts within a few minutes (**Figure 1A** and **1B**, Videos 1 and 2). Dual detection of the chloroplast stromal marker RBCS fused to enhanced yellow fluorescent protein (RBCS-EYFP) and of chlorophyll autofluorescence confirmed that the budding structure specifically contained stroma material (EYFP positive) without any chlorophyll signal, a marker of thylakoid membrane (**Figure 1C**, Video 3). These time-lapse observations reveal that chloroplasts form budding structures that are released in sugar-starved leaves. The chloroplast buds are cytoplasmic spherical bodies (approximately 0.5 to 1.5 μm in diameter) that are labeled by a chloroplast stroma marker and are adjacent to a main chloroplast body.

To estimate the impact of the release of chloroplast buds on the decline in chloroplast volume, we measured the volume of RBCS-mRFP signals in wild-type (WT) and *atg7* leaves before and after incubation in the dark for 24 h (**Figure 1-figure supplement 2**). Chloroplasts accumulate starch during the day and consume most of this starch during the night (Smith and Stitt, 2007). To minimize the effects of starch metabolism on the changes in chloroplast volume, we began sugar starvation treatment at the end of the night when a major portion of the starch has been consumed. After the treatment, stroma volume decreased in both genotypes (**Figure 1-figure supplement 2B**). However, the decline in volume was smaller in *atg7* leaves (14%) than in WT leaves (30%). The vacuolar transport of chloroplast budding structures as RCBs did not occur in *atg7* leaves (**Figure 1-figure supplement 1**). These results suggest that the release of chloroplast buds contributes to the decrease in chloroplast stroma volume in sugar-starved leaves.

We then performed a simultaneous detection of three organelle markers using conventional confocal microscopes to determine if the chloroplast buds contained membrane-bound proteins. Accordingly, we incubated leaves from plants accumulating the chloroplast outer envelope marker TRANSLOCON AT THE OUTER MEMBRANE OF CHLOROPLASTS 64 (TOC64) fused to mRFP and the stromal marker RBCS-GFP in darkness as described above. Under these conditions, we observed TOC64-mRFP signal surrounding a chloroplast bud (**Figure 2A**, arrowheads, **Figure 2-figure supplement 1A**). We then monitored chloroplasts in leaves accumulating the chloroplast inner envelope membrane protein K^+ EFFLUX ANTIporter 1 (KEA1) fused to mRFP along with RBCS-GFP after dark incubation. The chloroplast buds contained KEA1-mRFP signal but no chlorophyll signal (**Figure 2-figure supplement 1B**). We also captured the transport of an RCB with TOC64-mRFP or KEA1-mRFP signal (**Figure 2B**, Video 4, **Figure 2-figure supplement 2**, Video 5).

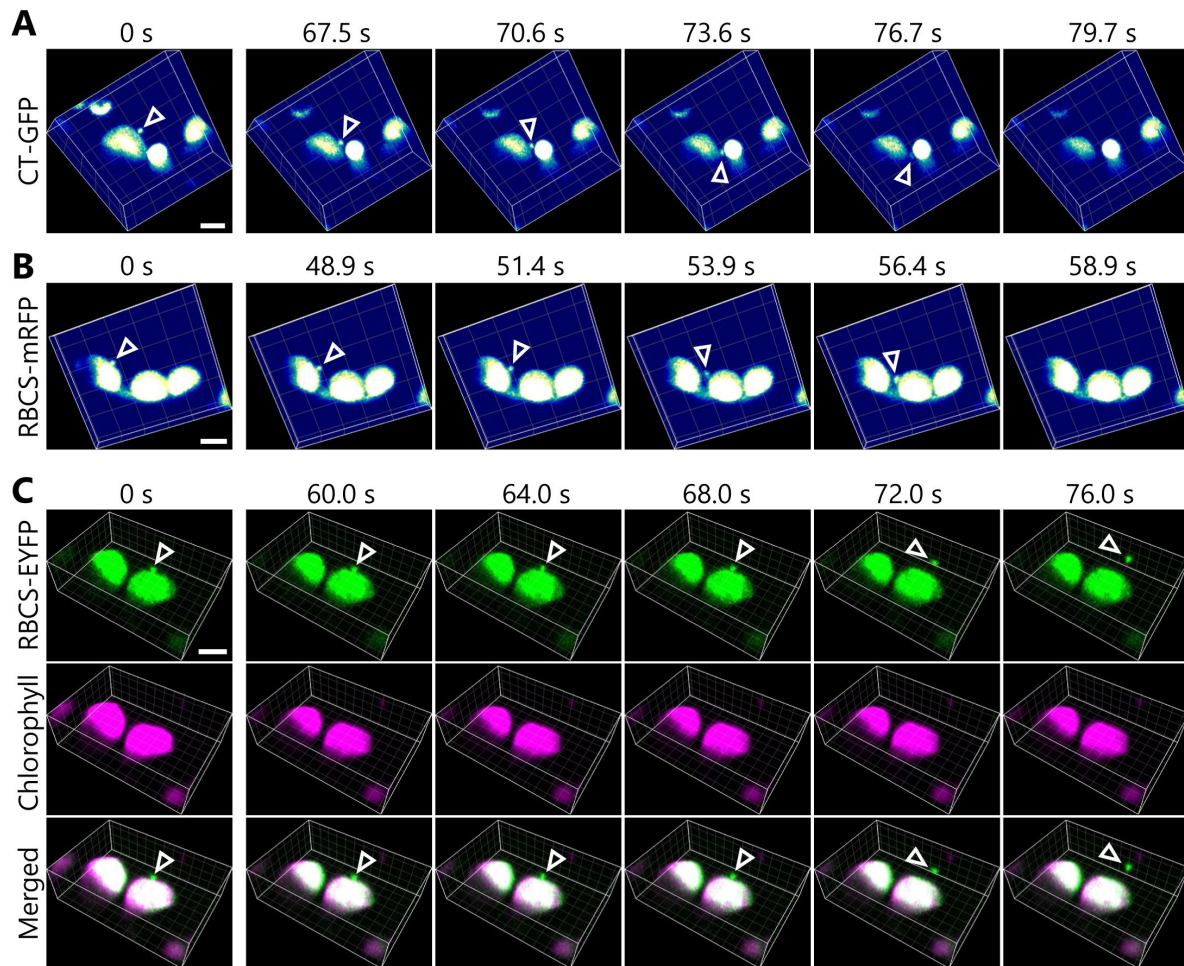


Figure 1.

Chloroplast buds are released in sugar-starved leaves.

Time-lapse observations of 3D-reconstructed chloroplast morphology in Arabidopsis mesophyll cells accumulating chloroplast stroma-targeted fluorescent markers. A leaf from a plant accumulating chloroplast stroma-targeted GFP (CT-GFP) (A), RBCS-mRFP (B), or RBCS-EYFP (C) was incubated in sugar-free solution in darkness for 5 h from dawn (A), 21 h (B), or 24 h from the period in light (C) and then observed through a two-photon excitation microscope equipped with a confocal spinning-disk unit. Second rosette leaves from 20-to 22-d-old plants were used. Images in (A–C) are still frames from Videos 1, 2, and 3, respectively. Time scales above the images indicate the elapsed time from the start of the respective videos. Arrowheads indicate chloroplast budding structures. Scale bars, 5 μm. In (C), green, RBCS-EYFP; magenta, chlorophyll fluorescence. In the merged images, the overlapping regions of RBCS-EYFP and chlorophyll signals appear white.

These results indicate that the budding structures contained chloroplast envelope components. In another set of experiments, we generated transgenic plants accumulating the thylakoid membrane marker ATP synthase gamma subunit (ATPC1) fused to tagRFP and stromal RBCS-GFP. The ATPC1-tagRFP fluorescent signal fully overlapped with that of chlorophyll autofluorescence, with both originating from the thylakoid membrane (**Figure 2C** [↗](#) and **2D** [↗](#)). In the chloroplast budding structures, however, we detected no ATPC1-tagRFP signal, in contrast to the strong stromal RBCS-GFP signal (**Figure 2C** [↗](#), arrowheads, **Figure 2-figure supplement 1C** [↗](#)). In agreement with this observation, we captured a released RCB that did not contain any ATPC1-tagRFP or chlorophyll signals, but accumulated plentiful RBCS-GFP (**Figure 2D** [↗](#), Video 6). These observations support the notion that the chloroplast buds containing stromal and envelope proteins are released as RCBs without thylakoid membranes. These live-cell imaging results are consistent with an immuno-electron microscopy study of RCBs in wheat (*Triticum aestivum*) leaves (Chiba et al., 2003 [↗](#)).

The vacuolar membrane dynamically interacts with chloroplast stromal components to incorporate them into the vacuolar lumen

We tracked the trafficking of RCBs in mesophyll cells further and obtained a two-dimensional (2D) sequential series of images of the release and transport of an RCB in leaves accumulating stromal CT-GFP (**Figure 3A** [↗](#), Video 7). This time series included two phases during which the punctum moved quickly (**Figure 3B** [↗](#), 24.4–34.8 s and 73.8–90.5 s). A tracking analysis for the RCB indicated that during the early phase (24.4–34.8 s), the punctum moved rather smoothly in one general direction (**Figure 3C** [↗](#), Video 7); however, during the later phase (73.8–90.5 s), the punctum appeared to move more randomly, akin to Brownian movement (**Figure 3C** [↗](#), Video 7). Other tracking series of dark-incubated leaves accumulating RBCS-tagRFP or RBCS-EYFP also showed RCBs starting to exhibit random movement: after 185 s in **Figure 3-figure supplement 1A** [↗](#) (Video 8) or after 25 s in **Figure 3-figure supplement 1B** [↗](#) (Video 9). Previous studies showed that autophagic bodies accumulating into the vacuolar lumen exhibit such random movements (Ishida et al., 2008 [↗](#)). We thus reasoned that the incorporation of RCBs into the vacuolar lumen from the cytoplasm takes place when these structures start to exhibit random movement.

We therefore turned our attention to the dynamics of the vacuolar membrane when it is incorporating RCBs into the vacuolar lumen in transgenic plants accumulating the vacuolar membrane marker VACUOLAR H⁺-PYROPHOSPHATASE 1 (VHP1) fused to mGFP (Segami et al., 2014 [↗](#)) along with the stromal marker RBCS-mRFP (**Figure 4** [↗](#)). From one of our time-lapse imaging series, we identified still frames of an RCB present in the cytoplasm being engulfed by the vacuolar membrane (**Figure 4A** [↗](#), open arrowheads from 25.4–30.5 s, Video 10). Later in the time series, we observed the opening of the part of the vacuolar membrane that had engulfed the RCB, resulting in the RCB being released into the vacuolar lumen (**Figure 4A** [↗](#), filled arrowheads from 47.0–52.0 s, Video 10). We captured another instance of the dynamics of the vacuolar membrane in another time series, with the vacuolar membrane first surrounding the RCB (52.6–57.4 s) before releasing it (65.3–72.9 s) into the vacuolar lumen (**Figure 4B** [↗](#), Video 11). **Figure 4-figure supplement 1** [↗](#) and Videos 12–14 show similar membrane dynamics that were observed in three individual plants. An RCB marked by RBCS-mRFP was engulfed by the vacuolar membrane at 48.0–54.0 s (**Figure 4-figure supplement 1A** [↗](#), Video 12), 6.0–15.0 s (**Figure 4** [↗](#)–figure supplement 1B, Video 13), and 111.0–123.0 s (**Figure 4-figure supplement 1C** [↗](#), Video 14). The RCBs were then released into the vacuolar lumen. Our time-lapse imaging approach thus allowed us to successfully visualize the cytoplasm-to-vacuole transport progression of chloroplast stroma material via autophagy.

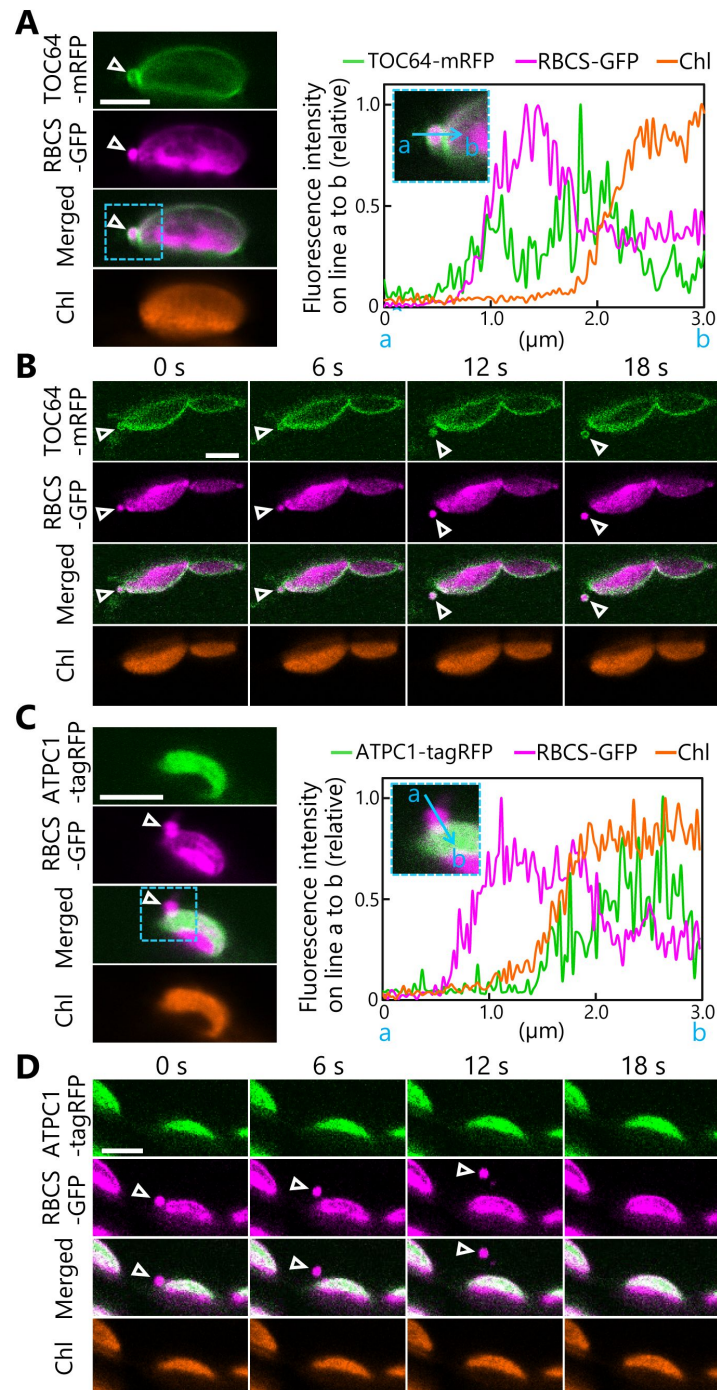


Figure 2.

Chloroplast buds containing stroma and envelope components are released from the chloroplasts.

Time-lapse observations of *Arabidopsis* mesophyll cells accumulating the chloroplast stroma marker along with an envelope marker or a thylakoid membrane marker. Leaves accumulating stromal RBCS-GFP along with outer envelope-bound TOC64-mRFP (A and B) or with thylakoid membrane-bound ATPC1-tagRFP (C and D) were incubated in sugar-free solution in darkness for 5 to 9 h from dawn and then observed. Second rosette leaves from 21-to 22-d-old plants were used. Images in (B) or (D) are still frames from Videos 4 and 6, respectively. Time scales above the images indicate the elapsed time from the start of the respective videos. Arrowheads indicate chloroplast budding structures. Scale bars, 5 μm. Green, TOC64-mRFP or ATPC1-tagRFP; magenta, RBCS-GFP; orange, chlorophyll (Chl) fluorescence. The graphs in (A and C) show fluorescence intensities along the blue lines (a to b) in the magnified images of the area indicated by dashed blue boxes. The intensities are shown relative to the maximum intensity for each fluorescence channel, set to 1.

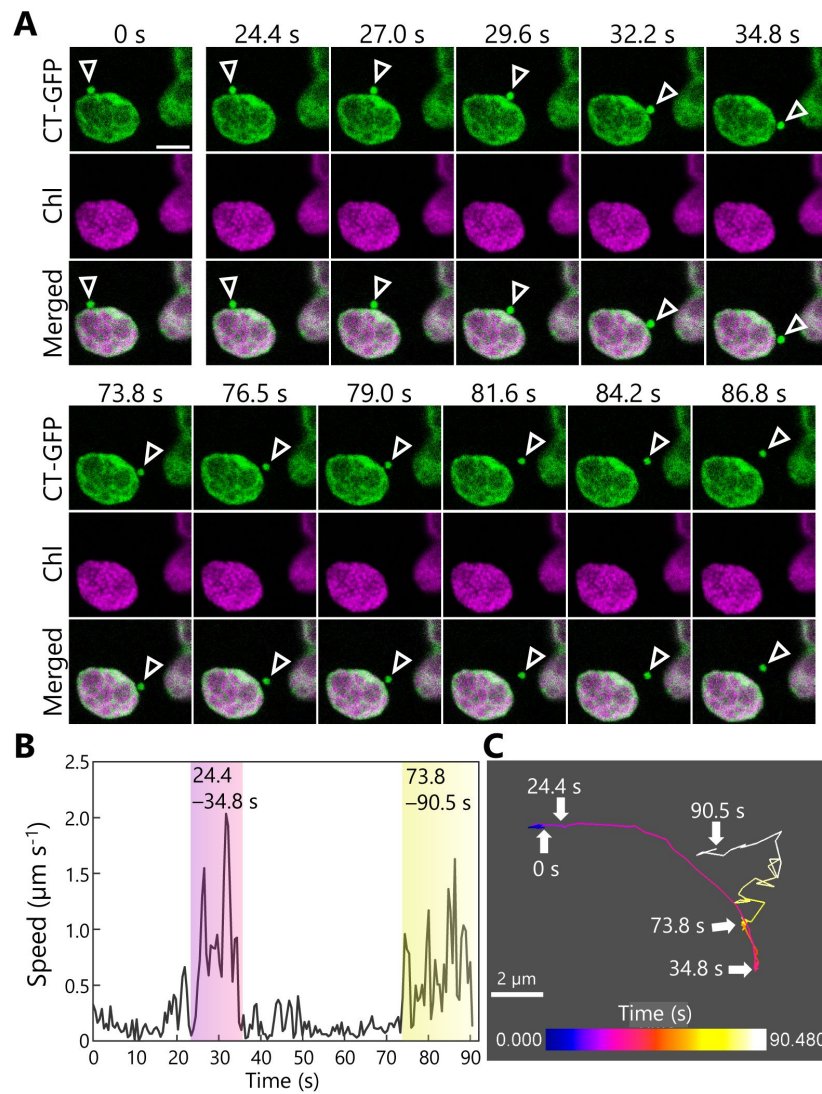


Figure 3.

Tracking the transport of a Rubisco-containing body.

A leaf accumulating chloroplast stroma-targeted GFP (CT-GFP) was incubated in sugar-free solution in darkness for 6 h from dawn, and the transport of a Rubisco-containing body (RCB) was tracked. The second rosette leaf from a 21-d-old plant was used. (A) Confocal images during the periods when the RCB moved quickly (24.4–34.8 and 73.8–90.5 s). Arrowheads indicate an RCB. The images are still frames from Video 7. Time scales above the images indicate the elapsed time from the start of the video. Green, CT-GFP; magenta, chlorophyll (Chl) fluorescence. Scale bar, 5 μm . (B) Calculated speed of the tracked RCB in (A). (C) The track of the RCB. The color of the track line changes over time, as indicated by the color bar. Scale bar, 2 μm .

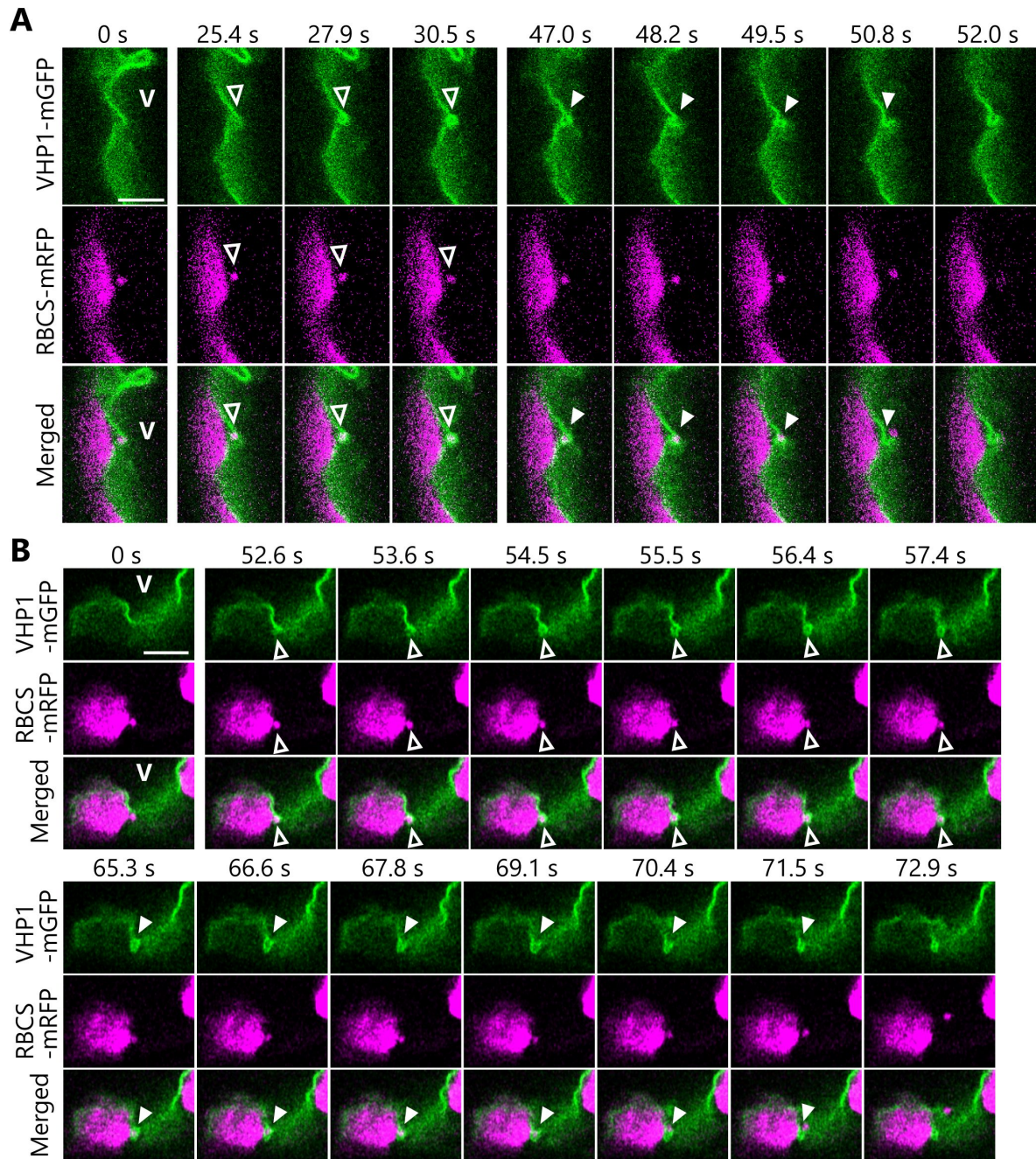


Figure 4.

Dynamics of the vacuolar membrane during the incorporation of Rubisco-containing bodies.

Leaves accumulating the chloroplast stroma marker RBCS-mRFP along with the vacuolar membrane marker VHP1-mGFP were incubated in sugar-free solution in darkness for 6 to 8 h from dawn, and the behavior of cytosolic RCBs was monitored. Second rosette leaves from 21- or 23-d-old plants were used. The images when the vacuolar membrane engulfs an RCB (25.4–30.5 s in A and 52.6–57.4 s in B) and when an RCB is incorporated into the vacuolar lumen (47.0–52.0 s in A and 65.3–72.9 s in B) are shown. The images in (A) and (B) are still frames from Videos 10 and 11, respectively. Time scales above the images indicate the elapsed time from the start of the respective videos. Open arrowheads indicate an RCB engulfed by the vacuolar membrane. Closed arrowheads indicate the open site of the vacuolar membrane for the incorporation of an RCB. V indicates the region of the vacuolar lumen. Green, VHP1-mGFP; magenta, RBCS-mRFP. Scale bars, 5 μ m.

Formation and segmentation of chloroplast buds occur simultaneously with autophagosome maturation

Based on the observation of spherical structures budding off from chloroplasts (**Figures 1** [↗](#)–**3** [↗](#)), we hypothesized that these structures are formed in response to sugar starvation and are subsequently recognized and sequestered by the autophagosomal membranes. We thus expected the leaves of autophagy-deficient mutants to accumulate multiple extended structures of chloroplast stroma as chloroplast protrusions upon exposure to sugar starvation. This hypothesis seemed consistent with the observations of stromule-rich cells in hypocotyls or dark-incubated leaves of autophagy-deficient *atg5* or *atg7* mutants, compared with non-mutants (Ishida et al., 2008 [↗](#); Spitzer et al., 2015 [↗](#)), assuming that the accumulation of chloroplast protrusions precedes stromule formation in *atg* mutants. To test this hypothesis, we subjected the leaves of *atg5* and *atg7* mutants, each defective in the function of a core ATG protein, to darkness-induced sugar starvation and treatment with concA for 1 d (**Figure 5** [↗](#)). We then evaluated the appearance of RCBs and chloroplast protrusions comprising extended structures of chloroplast stroma labeled by RBCS-mRFP, approximately 0.5 to 1.5 μm long. We detected no RCB in the leaves of either mutant (**Figure 5A** [↗](#) and **5B** [↗](#)); however, in both the WT and the mutants, we noticed fewer chloroplast protrusions in dark-incubated leaves treated with concA compared with the respective untreated control leaves, contrary to our hypothesis (**Figure 5C** [↗](#) and **5D** [↗](#)). We checked single mutants of *ATG2* and *ATG10* (*atg2* and *atg10*) to investigate other core ATGs and obtained similar results (**Figure 5–figure supplement 1** [↗](#)): The number of chloroplast protrusions did not increase in *atg2* or *atg10* leaves during a 1-d incubation in darkness.

We therefore posited that autophagosome formation might be required for the production of chloroplast buds. To test this idea, we observed the behavior of autophagosomes in transgenic plants accumulating the isolation membrane marker AUTOPHAGY8a (ATG8a) fused to GFP (as GFP-ATG8a) along with stromal RBCS-mRFP or chloroplast stroma-targeted (CT)-DsRed (**Figure 6** [↗](#)). Following incubation of leaves in darkness and confocal microscopy observation, we observed isolation membranes (marked by GFP-ATG8a) that were tightly associated with the chloroplast surface before any budding structure was visible (**Figure 6A** [↗](#) and **6B** [↗](#), 0 s). Over time, however, fluorescence monitoring of the same area revealed that the isolation membrane-associated site from the chloroplast side gradually protruded and eventually formed a spherical body containing RBCS-mRFP or CT-DsRed and surrounded by GFP-ATG8a signal, i.e., an autophagosome containing an RCB (**Figure 6A** [↗](#) and **6B** [↗](#), arrowheads; Videos 15 and 16).

Using the time-series data shown in **Figure 6B** [↗](#) (Video 16), we evaluated the time-dependent changes of the size of the chloroplast bud and the ratio between the major axis and the minor axis of the GFP-ATG8a-labeled autophagosome (**Figure 6C** [↗](#)). This ratio dropped to 1 as the isolation membrane became a spherical autophagosome. During this observation period, the ratio between the major and minor axes started to decrease at around 40 s and approached 1 at around 100 s. The chloroplast budding structure appeared at around 50 s and gradually increased in size until around 120 s. These data support the notion that the chloroplast bud appears as the isolation membrane develops and becomes a segmented RCB when the spherical autophagosome forms. We identified another example in an independent image series (**Figure 6** [↗](#)–figure supplement 1, Video 17) where the ratio between the major and minor axes of the isolation membrane started to decrease at approximately 40 s and the chloroplast bud appeared at around 50 s. In the third set of images, we observed the formation of chloroplast buds and later of RCBs near GFP-ATG8a-associated sites on multiple chloroplasts and sequentially (**Figure 6–figure supplement 2** [↗](#), Video 18). Therefore, we conclude that the development of the chloroplast-associated isolation membrane precedes the production of chloroplast buds and of RCBs for piecemeal-type chloroplast autophagy.

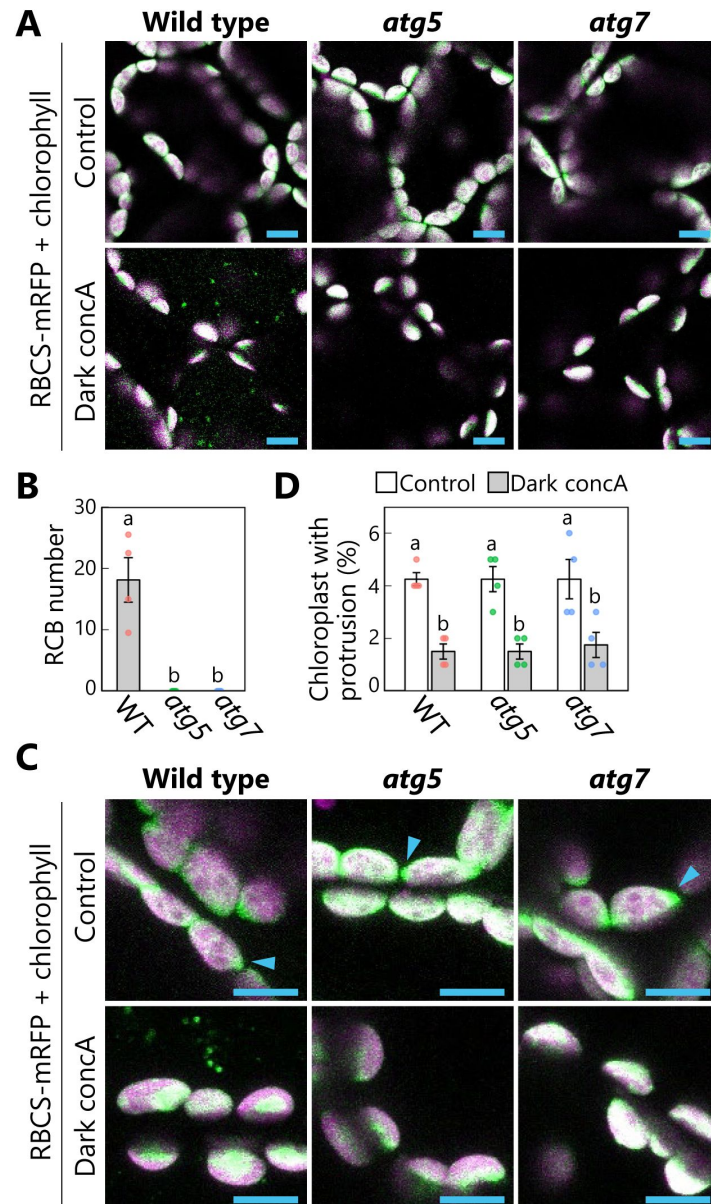


Figure 5.

Autophagy deficiency does not increase the number of chloroplast protrusions during a 1-d dark treatment.

Leaves from wild-type (WT), *atg5*, or *atg7* plants accumulating the chloroplast stroma marker RBCS-mRFP were incubated in sugar-free solution containing 1 μ M concanamycin A (concA) for 1 d in darkness. Second rosette leaves from 21-d-old plants were used. 2D images of mesophyll cells were acquired (A), and the number of accumulated RCBs in the vacuoles was scored (B). Small puncta containing RBCS-mRFP without the chlorophyll signal appear green and are RCBs in the vacuole. Leaves from untreated plants are shown as control. The appearance of chloroplast protrusions was observed from orthogonal projections created from z-stack images (15 μ m in depth; C), and the proportion of chloroplasts forming protrusion structures out of 50 chloroplasts was scored (D). Scale bars, 10 μ m. Green, RBCS-mRFP; magenta, chlorophyll fluorescence. Only the merged channels are shown. The overlapping regions of RBCS-mRFP and chlorophyll signals appear white. Arrowheads indicate the structures that were counted as a chloroplast protrusion in (D). Different lowercase letters denote significant differences based on Tukey's test ($P < 0.05$). Values are means $\pm$ SE ($n = 4$). Dots represent individual data points in each graph.

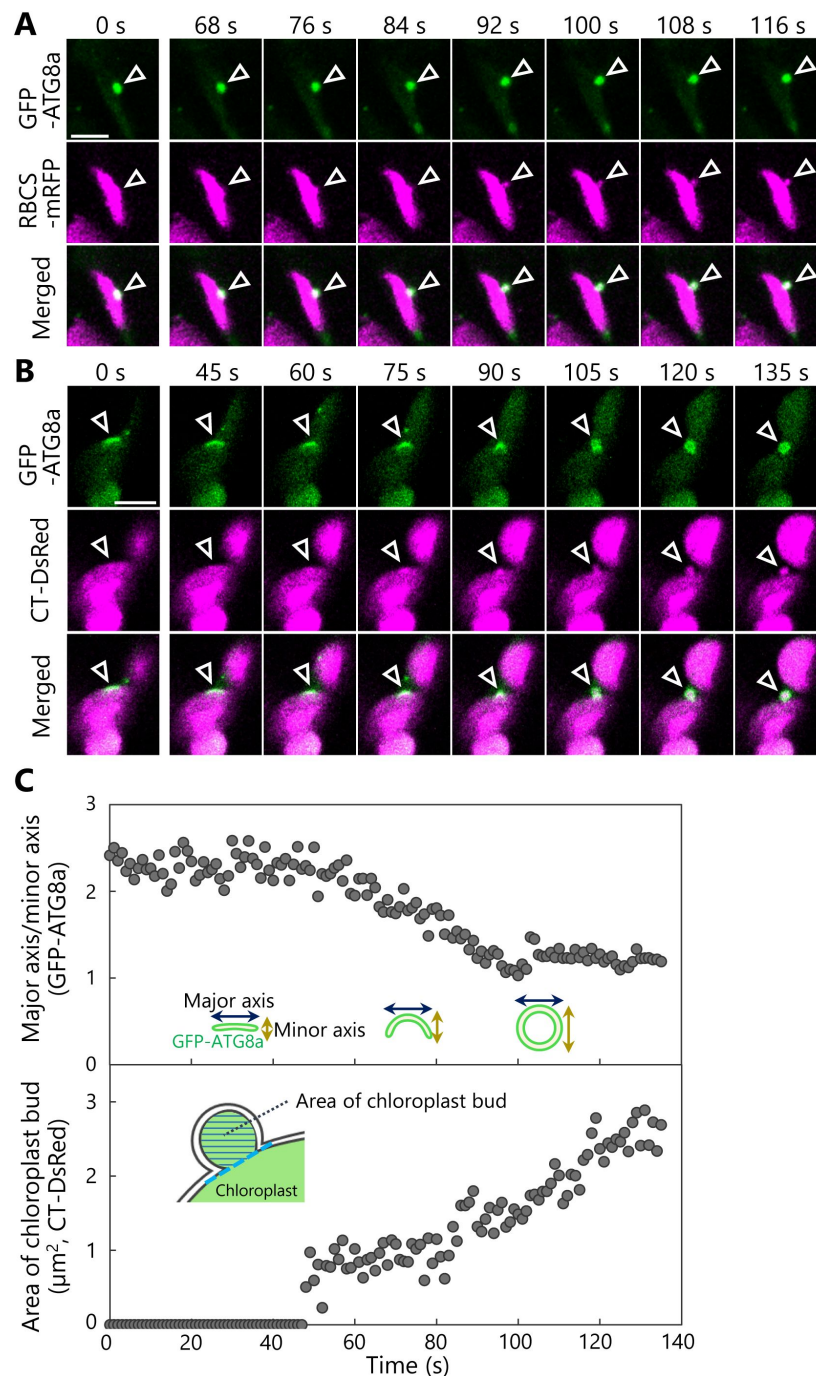


Figure 6.

The formation of a chloroplast bud and the maturation of the chloroplast-associated isolation membrane occur concomitantly.

Leaves accumulating the chloroplast stroma marker, RBCS-mRFP (A) or CT-DsRed (B), and the isolation membrane marker GFP-ATG8a were incubated in sugar-free solution in darkness for 6 h from dawn (A) or 21 h from the period in light (B) and then observed. Second rosette leaves from 23-d-old plants were used. Arrowheads indicate the position of the chloroplast-associated isolation membrane. Images in (A) and (B) are still frames from Videos 15 and 16, respectively. Time scales above the images indicate the elapsed time from the start of the respective videos. Green, GFP-ATG8a; magenta, RBCS-mRFP or CT-DsRed. Scale bars, 5 μm . (C) Time-dependent changes in the ratio of the major axis to the minor axis in the GFP-ATG8a-labeled isolation membrane (top) or in the area of the chloroplast budding structure (bottom) as measured from the images in (B).

We then evaluated the proportion of chloroplast buds associated with GFP-ATG8a signal when leaves accumulating RBCS-mRFP and GFP-ATG8a were incubated in the dark for 5 to 9 h from the end of night (**Figure 6-figure supplement 3** [↗](#)). Of the chloroplast buds examined, 64% were GFP-ATG8a positive (**Figure 6-figure supplement 3A** [↗](#)–**3C** [↗](#)). The formation of GFP-ATG8a-labeled chloroplast buds increased in response to the dark treatment (**Figure 6-figure supplement 3D** [↗](#)), but the buds did not appear in leaves from *atg7* plants exposed to dark treatment (**Figure 6-figure supplement 3E** [↗](#)). Therefore, a large portion of chloroplast buds may form synchronously with the development of the isolation membrane under sugar-starvation treatment. We then turned our attention to the isolation membrane and evaluated the proportion of GFP-ATG8a-labeled structures associated with chloroplasts (**Figure 6-figure supplement 4** [↗](#)). Of the isolation membranes examined, 23% localized on the chloroplast surface as flattened structures (**Figure 6-figure supplement 4A** [↗](#) and **4D** [↗](#)), and 20% localized with chloroplast buds as spherical or curved structures (**Figure 6-figure supplement 4B** [↗](#) and **4D** [↗](#)). Therefore, 43% of the GFP-ATG8a-labeled structures participated in chloroplast degradation during the dark treatment in this study. These results support the notion that chloroplast budding accompanied by autophagosome development substantially contributes to the degradation of chloroplast components in sugar-starved leaves.

SA influences stromule formation in autophagy-deficient mutants

Our initial hypothesis was that stromules are related to RCBs: Stromule-rich cells appeared because of impaired RCB production in *atg* mutants. However, our imaging assays above revealed that RCBs emerge from mesophyll chloroplasts in the absence of prior formation of stromules or protrusions (**Figure 6** [↗](#)), pointing to the existence of another factor responsible for the elevated stromule formation seen in *atg* mutant leaves. A previous study reported the hyperaccumulation of SA in senescing Arabidopsis *atg* mutants (Yoshimoto et al., 2009 [↗](#)). Since SA signaling stimulates immune responses, including stromule formation (Caplan et al., 2015 [↗](#)), we speculated that SA might contribute to stromule formation in autophagy-deficient mutants. To investigate this possibility, we generated transgenic plants in the *atg5* mutant background accumulating stromal GFP (CT-GFP) and a construct carrying NahG, encoding a bacterial SA hydroxylase that catabolizes SA (Delaney, 1994; Yoshimoto et al., 2009 [↗](#)). In these plant lines, we evaluated the frequency of chloroplasts forming stromules, which are thin, tubular structures of chloroplast stroma less than 1 μ m in diameter (Hanson and Sattarzadeh, 2011 [↗](#); Brunkard et al., 2015 [↗](#)). We examined chloroplasts in guard cells, as this cell type actively forms many more stromules than the mature chloroplasts of mesophyll cells (Ishida and Yoshimoto, 2008 [↗](#)). Chloroplasts in the guard cells of the *atg5* mutant formed more stromules than WT and NahG-expressing plants (**Figure 7A** [↗](#) and **7B** [↗](#)). The introduction of the NahG construct into the *atg5* mutant background (*atg5* NahG) largely abolished stromule formation (**Figure 7A** [↗](#) and **7B** [↗](#)). To further explore the role of SA in stromule formation, we generated double mutants of *SALICYLIC ACID INDUCTION DEFICIENT 2* (*SID2*), encoding the SA biosynthetic enzyme ISOCHORISMATE SYNTHASE 1, and *ATG5* or *ATG7* (*sid2 atg5* and *sid2 atg7*). The *sid2* mutation also resulted in fewer stromules in the double mutant relative to the respective *atg5* and *atg7* single mutants in guard cells (**Figure 7C** [↗](#) and **7D** [↗](#)).

We next observed stromule formation in mesophyll chloroplasts accumulating CT-GFP or RBCS-mRFP and found that many chloroplasts formed stromules in senescent leaves of 36-d-old *atg5* and *atg7* plants (**Figure 7E** [↗](#) and **7F** [↗](#), **Figure 7-figure supplement 1** [↗](#)). In cells from the third rosette leaves of 36-d-old plants accumulating CT-GFP, we observed stromules in 22.5% or 15.3% of chloroplasts from the *atg5* or *atg7* single mutants, respectively; however, the introduction of the *sid2* mutation in *sid2 atg5* and *sid2 atg7* led to fewer stromules (9.8% or 2.5% in *sid2 atg5* or *sid2 atg7*, respectively; **Figure 7E** [↗](#) and **7F** [↗](#)). In the leaves of 20-d-old plants, we did not detect stromules in any genotype (**Figure 7-figure supplement 2** [↗](#)). These results support the notion that SA accumulation due to autophagy deficiency activates stromule formation during leaf senescence.

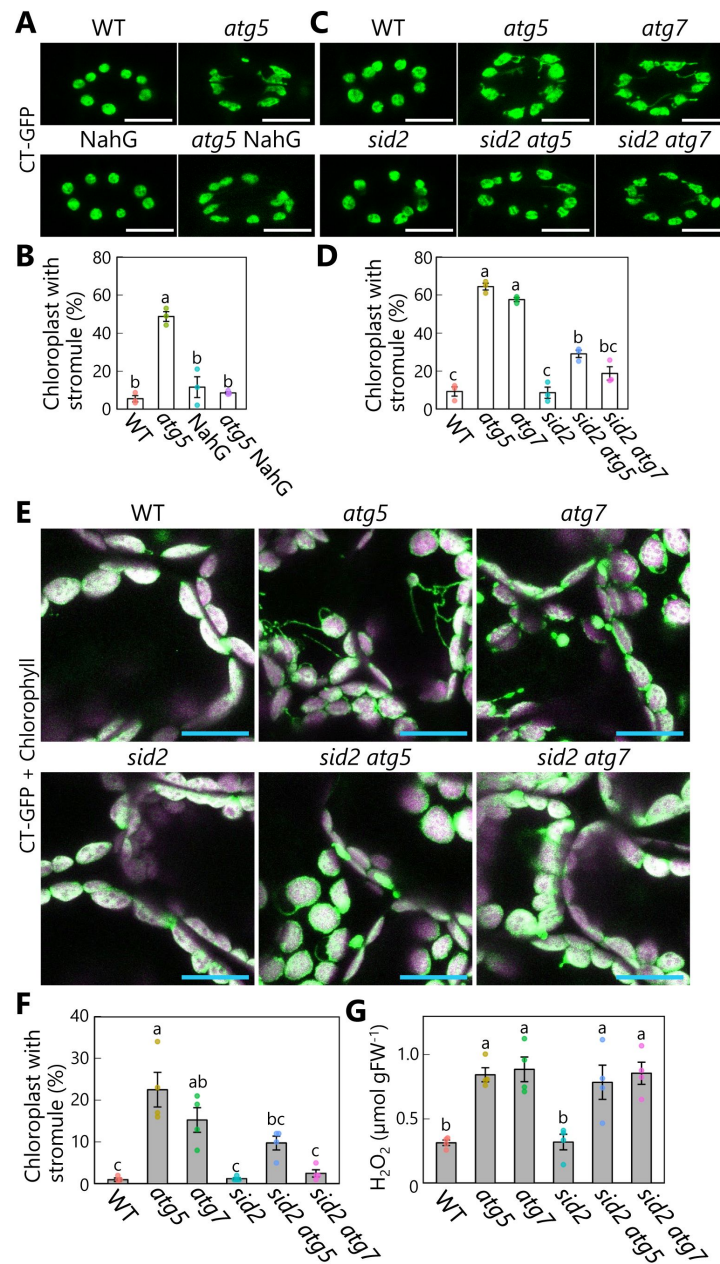


Figure 7.

Diminished salicylic acid signal suppresses stomule formation in autophagy-deficient mutants.

(A) Orthogonal projections produced from z-stack images (10 μm in depth) of guard cells from wild-type (WT), *atg5*, *NahG*, and *atg5 NahG* leaves accumulating chloroplast stroma-targeted GFP (CT-GFP). Second rosette leaves from 13-d-old plants were used. Scale bars, 10 μm. (B) Percentage of chloroplasts forming stromules in 10 stomata, from the observations described in (A). (C) Orthogonal projections produced from z-stack images (10 μm in depth) of guard cells from WT, *atg5*, *atg7*, *sid2*, *sid2 atg5*, and *sid2 atg7* leaves accumulating CT-GFP. Scale bars, 10 μm. (D) Percentage of chloroplasts forming stromules in 10 stomata, from the observations described in (C). (E) Orthogonal projections produced from z-stack images (15 μm in depth) of mesophyll cells from WT, *atg5*, *atg7*, *sid2*, *sid2 atg5*, and *sid2 atg7* leaves accumulating CT-GFP. Third rosette leaves from 36-d-old plants were observed. Green, CT-GFP; magenta, chlorophyll fluorescence. Only the merged channels are shown. Scale bars, 20 μm. (F) Percentage of chloroplasts forming stromules out of 50 chloroplasts in mesophyll cells, from the observations described in (E). (G) Hydrogen peroxide (H₂O₂) content in third rosette leaves from 36-d-old WT, *atg5*, *atg7*, *sid2*, *sid2 atg5*, and *sid2 atg7* plants. Different lowercase letters denote significant differences based on Tukey's test ($P < 0.05$). Values are means ± SE ($n = 3$ in B and D or 4 in F and G). Dots represent individual data points in each graph.

The frequency of stromules was higher in the *sid2 atg5* and *sid2 atg7* double mutants than in WT plants and the *sid2* single mutant (**Figure 7**). We measured the content of hydrogen peroxide (H_2O_2) in all plants, since reactive oxygen species also trigger stromule formation (Caplan et al., 2015). In 36-d-old plants, *atg5* or *atg7* leaves contained 2.7 or 2.8 times more H_2O_2 than WT leaves (**Figure 7G**). Notably, the presence of the *sid2* mutation in the *sid2 atg5* and *sid2 atg7* double mutants did not alleviate H_2O_2 accumulation (**Figure 7G**). Such ROS accumulation may thus be one factor responsible for the greater number of stromules in *sid2 atg5* and *sid2 atg7* compared with WT and *sid2* plants. Moreover, NahG expression did not counteract the increase in H_2O_2 level caused by the *atg5* mutation (**Figure 7-figure supplement 3**). ROS accumulation in *atg* leaves may not depend on stimulation of SA signaling. These results are consistent with the results of histochemical ROS staining of Arabidopsis leaves in a previous study (Yoshimoto et al., 2009).

DRP5b is not required for autophagy of chloroplast fragments

Simultaneous progression of autophagosome formation and organelle segmentation has been previously observed during mitophagy in mammals and budding yeast (Yamashita et al., 2016; Fukuda et al., 2023). This autophagy-related mitochondrial division does not require the mammalian Drp1- or yeast Dnm1-dependent division machinery, respectively. Here, we examined whether the DRP5B-mediated chloroplast division participates in chloroplast autophagy and RCB production.

To this end, we generated *drp5b* mutant lines accumulating the stromal marker RBCS-mRFP or stromal GFP (CT-GFP). Although the mature leaves of *drp5b* had larger chloroplasts than WT leaves, reflective of the impaired chloroplast division during cell expansion typical of this mutant, many RCBs accumulated when the leaves were incubated in darkness in the presence of concA (**Figure 8A** and **8B**, **Figure 8-figure supplement 1**). We detected no RCB in the leaves of the *drp5b atg5* double mutant (**Figure 8-figure supplement 1**). These results indicate that RCB-mediated chloroplast autophagy is active in the *drp5b* mutant. To obtain an independent confirmation of this result, we performed an immunoblot assay of autophagy flux, based on the detection of free mRFP derived from the vacuolar cleavage of RBCS-mRFP (Ono et al., 2013). When we incubated the leaves of plants accumulating RBCS-mRFP in darkness without concA, free mRFP levels increased in leaves of WT but not *atg5* or *atg7* plants (**Figure 8C** and **8D**), indicating the occurrence of autophagy-dependent degradation of stromal proteins in response to sugar starvation in WT plants. The accumulation of free mRFP in the *drp5b* mutant was at least as high as that seen in WT plants (**Figure 8C**). The ratio of free mRFP to RBCS-mRFP increased in WT and *drp5b* plants after dark treatment (6.1 or 4.1 times higher compared with control conditions in WT plants or *drp5b*, respectively), consistent with autophagy-mediated degradation of chloroplast fragments (**Figure 8D**). Confocal microscopy observations of mesophyll cells also showed the spread of mRFP signal in the vacuolar lumen in dark-treated leaves of WT and *drp5b* plants (**Figure 8-figure supplement 2**). These results indicate that the autophagic degradation of chloroplast stroma is active in both genotypes. The accumulation of free mRFP was 3.2 times higher in untreated leaves of *drp5b* compared with untreated WT leaves (**Figure 8D**); therefore, the activity of chloroplast autophagy might be constitutively higher in leaves of *drp5b* than in those of WT plants.

We observed the budding off of chloroplast fragments in leaves of the *drp5b* mutant. We incubated a leaf from the *drp5b* mutant accumulating RBCS-mRFP in darkness, which allowed us to observe the formation of a chloroplast bud and its release (**Figure 9A**, arrowheads; Video 19). In a leaf of the *drp5b* mutant accumulating RBCS-mRFP and GFP-ATG8a, we detected the vesiculation of a chloroplast fragment starting at the site to which the isolation membranes, labeled by GFP-ATG8a, were tightly associated (**Figure 9B**). The isolation membranes were anchored at two sites of a large chloroplast in a leaf of the *drp5b* mutant (**Figure 9B**, white and blue arrowheads); the

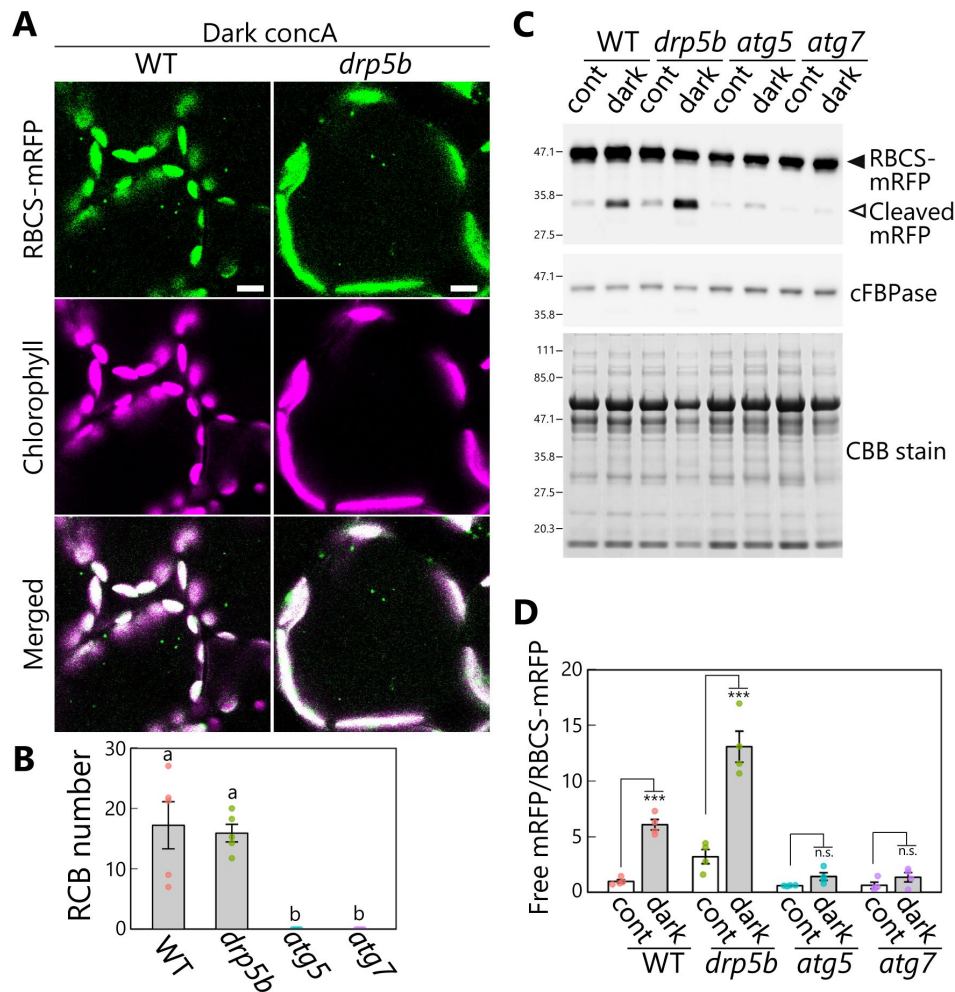


Figure 8.

DRP5b is dispensable for chloroplast autophagy in sugar-starved leaves.

(A) Confocal images of mesophyll cells from wild-type (WT) and *drp5b* leaves accumulating the stroma marker RBCS-mRFP. Second rosette leaves from 21-d-old plants were incubated in sugar-free solution containing 1 μ M concanamycin A (concA) in darkness for 1 d. Green, RBCS-mRFP; magenta, chlorophyll fluorescence. Scale bars, 10 μ m. (B) Number of accumulated RCBs in WT, *drp5b*, *atg5*, and *atg7* leaves, counted from the observations of leaves incubated in the dark with concA described in (A). Different lowercase letters denote significant differences based on Tukey's test ($P < 0.05$). Values are means $\pm$ SE ($n = 4-5$). (C) Biochemical detection of autophagy flux for chloroplast stroma based on a free RFP assay. RFP and cFBPase (loading control) were detected by immunoblotting of soluble protein extracts from leaves of WT, *drp5b*, *atg5*, and *atg7* plants accumulating RBCS-mRFP. Protein extracts from either untreated control leaves (cont) or leaves after 2 d of incubation in darkness (dark) were used. Total protein was detected by Coomassie Brilliant Blue (CBB) staining as a loading control. The filled arrowhead indicates RBCS-mRFP fusion, and the open arrowhead indicates free mRFP derived from the cleavage of RBCS-mRFP. (D) Quantification of the free mRFP/RBCS-mRFP ratio shown relative to that of untreated WT plants, which was set to 1. Asterisks denote significant differences based on t-test (***, $P < 0.001$; n.s., not significant). Values are means $\pm$ SE ($n = 4$). Dots represent individual data points in each graph.

chloroplast gradually protruded before releasing RCBs surrounded by autophagosomes (**Figure 9B** [↗](#), Video 20). We conclude that DRP5B is dispensable for the autophagy-related division of chloroplast fragments.

Discussion

Chloroplast division machinery functioning in piecemeal-type chloroplast autophagy

We previously demonstrated that chloroplast components in mesophyll cells are transported to the vacuole as a type of autophagic cargo termed RCBs to facilitate amino acid recycling ([Ishida et al., 2008](#) [↗](#); [Hirota et al., 2018](#) [↗](#)). However, how a portion of chloroplasts is mobilized into the vacuolar lumen remained uncertain. In this study, high-resolution time-lapse imaging techniques allowed us to visualize the trafficking progression of chloroplast fragments from their parental chloroplasts to the vacuolar lumen. Importantly, we established that the development of the chloroplast-associated isolation membrane and the division of the chloroplast fragment occur simultaneously. This autophagy-related chloroplast division does not rely on DRP5B-mediated chloroplast division. Likewise, the formation of chloroplast buds is unlikely to be linked to the formation of stromules. Therefore, an unknown division machinery may be required for the autophagy-related division of chloroplast fragments. A similar organelle division mechanism takes place during mitophagy in mammals and budding yeast. The site of a mitochondrion that is associated with the isolation membrane protrudes and divides as the isolation membrane develops and becomes an enclosed autophagosome ([Yamashita et al., 2016](#) [↗](#)). This type of mitochondrial division is independent of Drp1 and Dnm1. In yeast, the mitochondrial intermembrane-space protein mitofissin/Atg44 divides the mitochondrial fragment in coordination with autophagosome maturation ([Fukuda et al., 2023](#) [↗](#)). A functionally equivalent protein has not been identified in mammals.

Numerous studies have explored the molecular mechanisms of chloroplast division during the development of juvenile leaves ([Chen et al., 2018](#) [↗](#)). For instance, in Arabidopsis plants, ARC6 and PARALOG OF ARC6 (PARC6) mediate the tethering of the FTSZ ring on the chloroplast inner envelope ([Vitha et al., 2003](#) [↗](#)). PLASTID DIVISION 1 (PDV1) and PDV2 recruit DRP5B to the division site on the chloroplast outer envelope ([Miyagishima et al., 2006](#) [↗](#)). The ARC6–PDV2 and PARC6–PDV1 complexes may control the coordination of the stromal FTSZ ring and the cytosolic DRP5B ring ([Wang et al., 2017](#) [↗](#); [Chen et al., 2018](#) [↗](#)). Therefore, the controlled division of plastids requires the cooperation of multiple protein complexes across the inner and outer envelopes. Nevertheless, whether chloroplasts divide during their piecemeal degradation has not been evaluated. It seems likely that autophagy-related division of chloroplast fragments needs to be carefully regulated, as the size of the segment has to fit into the autophagosome. Since the volume occupied by chloroplasts per cell is tightly regulated during cell expansion ([Pyke and Leech, 1994](#) [↗](#)), the decline of chloroplast volume during starvation or senescence might also be highly controlled. Further studies are needed to elucidate the molecular machinery underlying the autophagy-related division of chloroplast fragments for degradation.

In our evaluation of chloroplast buds associated with GFP-ATG8a in excised leaves exposed to darkness (**Figure 6–figure supplement 3** [↗](#)), 36% of chloroplast budding structures were not labeled by GFP-ATG8a. Since endogenous ATG8 family members (ATG8a–8i) were functional, such structures might have formed due to the activities of non-labeled ATG8 proteins. Another possibility is that autophagy-independent chloroplast budding might occur. For example, stromule-related structures might form. CHLOROPLAST VESICULATION (CV)–containing vesicles are also observed as autophagy-independent puncta released from chloroplasts for degradation ([Wang and Blumwald, 2014](#) [↗](#); [Pan et al., 2023](#) [↗](#)). However, biochemical assays for chloroplast autophagy flux suggested that ATG-dependent pathways were the main routes for the vacuolar

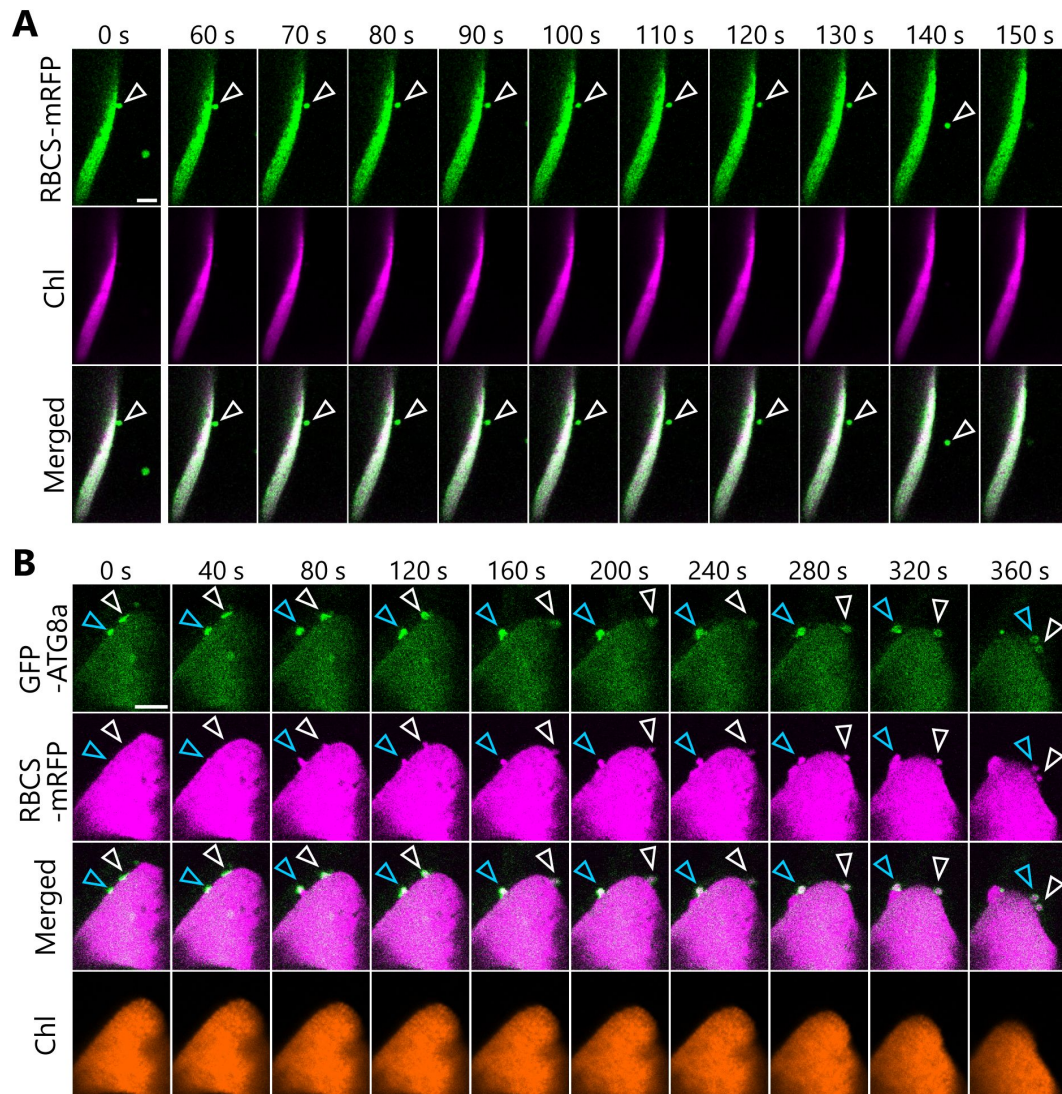


Figure 9.

Formation and segmentation of chloroplast buds in leaves of the *drp5b* mutant.

(A) A leaf from the *drp5b* mutant accumulating the stroma marker RBCS-mRFP was incubated in sugar-free solution in darkness for 5 h from dawn and then observed. Arrowheads indicate a chloroplast bud. Green, RBCS-mRFP; magenta, chlorophyll fluorescence (Chl). (B) A leaf from the *drp5b* mutant accumulating the stroma marker RBCS-mRFP and the isolation membrane marker GFP-ATG8a were incubated in sugar-free solution in darkness for 6 h from dawn and then observed. Second rosette leaves from 21- to 22-d-old plants were used. Arrowheads indicate the position of the chloroplast-associated isolation membrane. White and blue arrowheads indicate different isolation membrane-associated sites in a chloroplast. Images in (A) and (B) are still frames from Videos 19 and 20, respectively. Time scales above the images indicate the elapsed time from the start of the respective videos. Green, GFP-ATG8a; magenta, RBCS-mRFP; orange, chlorophyll (Chl) fluorescence. Scale bars, 5 μ m.

degradation of stromal components during the darkness-induced sugar starvation treatment used in this study (**Figure 8**). Previous studies have revealed that CV plays a role in plant responses to abiotic stress, such as drought or salt stress (Wang and Blumwald, 2014; Pan et al., 2023). A recent study suggested that autophagy, but not CV-dependent pathways, is the main contributor to the adaptive response of metabolism to sugar starvation (Barros et al., 2023). Overall, these insights support the notion that, during the early phase of sugar starvation (such as 1 to 2 d of extended darkness), the autophagy-associated machinery is mainly responsible for chloroplast segmentation and degradation. It is unclear what type of process contributed to the decrease in chloroplast stroma volume in leaves of *atg7* plants exposed to dark treatment (**Figure 1-figure supplement 2**).

An assay of autophagic flux of RBCS-mRFP protein in leaves from *drp5b* mutants suggested the possibility that chloroplast autophagy activity is constitutively higher in leaves of *drp5b* than in WT plants (**Figure 8**). The impaired chloroplast division due to the *drp5b* mutation might affect photosynthesis-mediated energy production and reduce energy availability, thereby activating chloroplast autophagy without dark treatment. As described in the Introduction, entire organelle-type autophagy (chlorophagy) mediates the degradation of chloroplasts in their entirety. Mesophyll cells in leaves of *drp5b* mutants contain few chloroplasts (approximately 4 per cell), whereas those in WT leaves contain approximately 60 chloroplasts (Miyagishima et al., 2006). Since the volume in a cell occupied by chloroplasts is similar between genotypes, chloroplasts from *drp5b* are much larger than WT chloroplasts. Chloroplasts that are too large might show inhibited chlorophagy, resulting in the alternative activation of piecemeal-type chloroplast autophagy. Analyzing how chloroplasts are degraded in mutants of chloroplast division ring components might provide insight into the interplay between entire organelle-type autophagy and piecemeal-type autophagy during chloroplast degradation.

Autophagy receptor and adapter proteins for the degradation of mature chloroplasts

Organelle-selective autophagy is typically controlled by receptor proteins that recognize the target organelles and act as a bridge between the organelles and isolation membrane-anchored ATG8 (Farre and Subramani, 2016). The observation of a chloroplast-associated isolation membrane (**Figure 6**) suggests that the receptors for chloroplast autophagy connect the isolation membrane to the chloroplast envelope. Such interaction might facilitate the localization of the isolation membrane on the region of subsequent chloroplast budding and its development along the chloroplast surface. In budding yeast, the accumulation of the mitophagy receptor Atg32 is required for the formation of the mitophagosome. Atg32 binds to the mitochondrial outer membrane and interacts with Atg8 and Atg11, the latter being a scaffold protein recruiting other ATG members, for autophagosome formation. In fission yeast (*Schizosaccharomyces pombe*), Atg43 located on the mitochondrion outer membrane binds to Atg8 to stabilize the autophagosomal membrane for mitophagosome formation during starvation (Fukuda et al., 2020). In mammalian mitophagy, five types of mitophagy receptors have been identified that bind to LC3 (light chain 3) or GABARAP proteins, the mammalian orthologs of ATG8 (Onishi et al., 2021). Whether similar receptors work for chloroplast autophagy in mature leaves has not been established. Atg39 is the receptor for nucleophagy in budding yeast (Mochida et al., 2015), during which the condensation of Atg39 enables the protrusion of the nuclear membrane to be sequestered by the autophagosome (Mochida et al., 2022). The accumulation of receptor proteins might contribute to chloroplast budding to form RCBs.

Another important regulator of the selective autophagy of organelles is the ubiquitination of the targets of this process. One well-characterized process is mammalian mitophagy regulated by PTEN-induced kinase 1 (PINK1) and Parkin (Ohnishi et al., 2021). PINK1 accumulates on the surfaces of damaged mitochondria, facilitating the ubiquitination of the mitochondria by the E3 ubiquitin ligase Parkin. Subsequently, autophagy adapter proteins connect the ubiquitin chain to

LC3/GABARAP proteins for the selective elimination of the ubiquitinated mitochondria via autophagosomes (Lazarou et al., 2015 [↗](#)). Five autophagy adaptor proteins have been identified in mammals: p62/Sequestosome-1 (SQSTM1), Next to BRCA1 gene 1 (NBR1), Tax1-binding protein 1 (TAX1BP1), nuclear dot protein52 (NDP52), and optineurin (OPTN). Among these, a plant ortholog of NBR1 has been identified (Svenning et al., 2011 [↗](#)). A recent study found that Arabidopsis NBR1 accumulates on damaged chloroplasts to induce their removal in their entirety (Lee et al., 2023 [↗](#)). NBR1 may participate in the recognition of damaged chloroplasts during the entire organelle-type degradation of chloroplasts by chlorophagy. Another study demonstrated that NBR1 regulates the selective degradation of the outer envelope-bound TOC protein complexes that control cytoplasm-to-plastid protein import (Wan et al., 2023 [↗](#)). TOC proteins are ubiquitinated under stress conditions such as ultraviolet-B exposure or high temperature and are then degraded by autophagosomes through NBR1-mediated recognition. These findings suggest that NBR1 plays an important role in the degradation of damaged chloroplast proteins. Whether ubiquitination and NBR1 participate in the degradation of portions of chloroplasts via RCBs under sugar-starvation conditions has not been evaluated.

Roles of stromules

The molecular functions of stromules are not completely understood. We predicted a functional link between stromules and RCBs such that stromule-associated structures are vesiculated and become RCBs for autophagic degradation. However, our imaging assays revealed that chloroplast stromule formation is not essential for RCB formation in mesophyll cells (**Figure 6** [↗](#)). The use of the *sid2* mutant and NahG transgenic lines indicated the close association between enhanced SA signaling and the elevated stromule formation observed in *atg* mutants (**Figure 7** [↗](#)). This finding is consistent with the suggested role of stromules in inducing programmed cell death as an SA-dependent immune response (Caplan et al., 2015 [↗](#)). We attribute the enhanced stromule formation in *atg* mutant leaves to the higher SA and ROS levels, alone or in combination with the excess stromal fraction accumulating in these mutants due to impaired RCB formation.

In root cells, an association of the isolation membrane with stromules of non-green plastids was reported (Spitzer et al., 2015 [↗](#)). Stromule formation is more common in the non-green plastids of tissues without photosynthetic activity than in the mature chloroplasts in leaf mesophyll cells (Ishida and Yoshimoto, 2008 [↗](#); Hanson and Sattarzadeh, 2008 [↗](#)). Therefore, the interaction between stromules and the isolation membrane might be important for the efficient degradation of non-green plastids by autophagy. Our study focused on the degradation mechanism of mature chloroplasts in leaves, since mature chloroplasts are rich in nutrients and amino acids and their degradation is particularly important for plant nutrient recycling (Makino and Osmond, 1991 [↗](#)). There might also be differences in the degradation mechanism of plastids among cell types.

Intracellular dynamics for the transport of RCBs

Soluble *N*-ethylmaleimide-sensitive factor attachment protein receptors (SNAREs) mediate versatile membrane fusion events (Ito and Uemura, 2022 [↗](#)). Multiple SNARE proteins participate in the fusion of the lysosomal/vacuolar membrane to the autophagosomal membrane in mammals and budding yeast. A recent study reported the involvement of the Arabidopsis SNARE proteins, VESICLE-ASSOCIATED MEMBRANE PROTEIN 724 (VAMP724) and VAMP726, in autophagosome formation (He et al., 2022 [↗](#)). In this study, we observed an interesting morphology of vacuolar membranes that incorporate RCBs into the vacuole (**Figure 4** [↗](#)). A similar phenomenon was previously observed during the transport of pexophagosomes (autophagosomes containing peroxisome components) in Arabidopsis leaves (Oikawa et al., 2022 [↗](#)). Thus, the transient engulfment of an autophagosome by the vacuolar membrane might be a common phenomenon for vacuolar incorporation in leaf mesophyll cells. However, the underlying mechanisms are unknown. In mammals, two autophagosomal membrane-anchored SNAREs, Syntaxin 17 (Stx17) and Ykt6, independently contribute to autophagosome-lysosome fusion (Itakura et al., 2012 [↗](#); Matsui et al., 2018 [↗](#)). Stx17 translocates to the closed autophagosome transiently, enabling its

fusion with the lysosome (Tsuboyama et al., 2016). Such a translocation system might allow the recruitment of the vacuolar membrane for its fusion with the mature autophagosomes containing RCBs in plant cells.

Concluding remarks

The current study revealed how a type of piecemeal autophagy transports chloroplast stroma and envelope components into the vacuole for degradation in mature *Arabidopsis* leaves. The key event is the development of the isolation membrane along the chloroplast surface, which may lead to the budding and segmentation of the membrane-contact site of a chloroplast. Proteins mediating chloroplast segmentation and the interaction between the chloroplast surface and the isolation membrane remain to be uncovered. This study developed a live-cell tracking method for piecemeal-type chloroplast autophagy, which will help future studies elucidate the underlying mechanisms of the intracellular dynamics of this type of autophagy.

Materials and Methods

Plant materials

Arabidopsis (*Arabidopsis thaliana*) plants from the Columbia accession (Col) were used in this study. Plants were grown in soil in growth chambers at 23°C under a 12-h-light/12-h-dark photoperiod with illumination from fluorescent lamps or LEDs (90–130 $\mu\text{mol m}^{-2} \text{s}^{-1}$). The T-DNA-insertion mutants for *ATG5* (*atg5-1*; SAIL_129_B07), *ATG7* (*atg7-2*; GABI_655B06), *ATG2* (*atg2-1*; SALK_076727), *ATG10* (*atg10-1*; SALK_084434), and *DRP5B* (*arc5-2*; SAIL_71_D11) have been described previously (Doelling et al., 2002; Thompson et al., 2005; Miyagishima et al., 2006; Phillips et al., 2008; Yoshimoto et al., 2009). The *sid2-2* mutant and a transgenic line expressing NahG have been reported previously (Delaney et al., 1994; Yoshimoto et al., 2009). The transgenic plants expressing a construct encoding chloroplast stroma-targeted GFP (GFP fused to the transit peptide of RECA protein) under the control of the cauliflower mosaic virus (CaMV) 35S promoter (*Pro35S:CT-GFP*; Kohler et al., 1997), chloroplast stroma-targeted DsRed (DsRed fused to the transit peptide of RECA protein) from the CaMV 35S promoter (*Pro35S:CT-DsRed*; Ishida et al., 2021), RBCS2B-GFP from the *Arabidopsis* *RBCS2B* promoter (*ProRBCS:RBCS-GFP*; Ishida et al., 2008), VHP1-mGFP from the *Arabidopsis* *VHP1* promoter (*ProVHP1:VHP1-mGFP*; Segami et al., 2014), GFP-ATG8a from the *Arabidopsis* *UBQ10* promoter (*ProUBQ:GFP-ATG8a*; Nakamura et al., 2021b), and TOC64-mRFP from the *Arabidopsis* *TOC64* promoter (*ProTOC64:TOC64-mRFP*; Kusano et al., 2023) have been reported in previous studies. Transgenic plants expressing *RBCS2B-EYFP*, *RBCS2B-mRFP*, or *RBCS2B-tagRFP* from the *RBCS2B* promoter (*ProRBCS:RBCS-EYFP*, *ProRBCS:RBCS-mRFP*, and *ProRBCS:RBCS-tagRFP*) were generated as follows. A genomic fragment encompassing the promoter region and the full-length coding region of *RBCS2B* (At5g38420), cloned into the pENTR/D/TOPO vector (Ishida et al., 2008), was inserted into the Gateway vectors pGWB540, pGWB553, or pGWB559 (Nakagawa et al., 2007) via LR clonase (Invitrogen) reaction to generate constructs encoding *RBCS2B-EYFP*, *RBCS2B-mRFP*, or *RBCS2B-tagRFP*, respectively.

Transgenic plants expressing *ATPC1-tagRFP* (*ProATPC1:ATPC1-tagRFP*) were produced as follows. A genomic fragment containing the promoter region and full-length coding region of *ATPC1* (At4g04640) was amplified from Col-0 genomic DNA by PCR using PrimeSTAR DNA polymerase (TaKaRa) and the primers ATPC1_F (CACCATGGAGAGGGCTCGTACCTTAC) and ATPC1_R (AACCTGTGCATTAGCTCCAG), cloned into pENTR/D-TOPO (Invitrogen), and then recombined into the vector pGWB559 via LR reaction. Transgenic plants expressing *KEA1-mRFP* (*ProKEA1:KEA1-mRFP*) were produced as follows. A genomic fragment containing the promoter region and full-length coding region of *KEA1* (At1g01790) was amplified from Col-0 genomic DNA by PCR using the primers KEA1_F (AGGAACCAATTAGTCTGACTCATGATCATAACAAGTCTC) and KEA1_R (AAAGCTGGGTCTAGATATCCGATTACGACTGTGCCTCTTC), cloned into pENTR1A (Invitrogen) using NEBuilder HiFi DNA Assembly master mix (New England Biolabs), and recombined into the vector

pGWB553 via LR reaction. The resulting construct was introduced into Arabidopsis plants by the floral dip method (Clough and Bent, 1998 [\[1\]](#)) using *Agrobacterium* (*Agrobacterium tumefaciens*) strain GV3101. Transgenic plants expressing two types of fluorescent markers and mutant plants expressing fluorescent markers were generated by crossing or additional transformations.

Live-cell imaging by confocal microscopy

The second rosette leaves of 20-to 24-d-old plants were used for time-lapse imaging of living cells. The leaves were excised, infiltrated with sugar-free solutions, and incubated in darkness for 5 to 24 h, followed by confocal microscopy observations. When the dark treatment was started at the end of the night, the incubation time was around 5–9 h. When the treatment was started during the light period, the incubation time was around 20–24 h. The sugar-free solutions contained 10 mM MES-NaOH, pH 5.5, alone or with full-strength Murashige and Skoog salts (Shiotani M. S.) or the photosynthesis inhibitor 3-(3,4-dichlorophenyl)-1,1-dimethylurea (DCMU).

Time-lapse imaging in **Figure 1** [\[1\]](#) was performed as previously described with a two-photon excitation confocal microscope with a spinning-disk unit (Otomo et al. 2015 [\[2\]](#)). The GFP signal was excited by 920-nm femtosecond light pulses generated by a mode-locked titanium-sapphire laser light source (Mai Tai eHP DeepSee; Spectra Physics). The YFP, RFP, and chlorophyll signals were excited by 1040-nm femtosecond light pulses generated by an ytterbium laser light source (femtoTrain; Spectra Physics). The fluorescent signals were observed under an inverted microscope (IX-71; Olympus) equipped with a spinning-disk scanner with 100- μ m-wide pinholes aligned with a Nipkow disk (CSU-MP ϕ 100; Yokogawa Electric) and a water-immersion lens (UPLSAPO60XW, numerical aperture [NA] = 1.20, Olympus). The fluorescence images were captured by an EM-CCD camera (EM-C2; Qimaging or iXon Ultra 897; Andor Technology) through a bandpass filter for GFP and YFP (BrightLine 528/38; Semrock), RFP (D630/60M; Chroma Technology), or chlorophyll autofluorescence (BrightLine 685/40; Semrock). For the simultaneous detection of YFP and chlorophyll signals, fluorescence was detected through image-splitting optics (W-View Gemini; Hamamatsu Photonics) including a dichroic mirror (FF580-FDi01-25 36; Semrock) and the bandpass filters. Z-Scans were performed with a piezo actuator (P-721; PI). The acquired images were processed and analyzed using NIS-Elements C software (Nikon) or Imaris software (Bitplane).

Time-lapse imaging analysis by confocal laser-scanning microscopy was performed with LSM800 (Carl Zeiss), LSM880 (Carl Zeiss), LSM900 (Carl Zeiss), or SP8 (Leica) systems. A water-immersion objective lens (C-Apochromat 40x, NA = 1.2 or LD C-Apochromat 63x, NA = 1.15; Carl Zeiss) or an oil-immersion objective lens (HC PL APO 63x, NA = 1.40, Leica) was used. Fast Airyscan mode was used for the observation displayed in **Figure 4B** [\[1\]](#) with the LSM880 system.

Quantification of microscopy images

Confocal images used for the quantification of RCB numbers (**Figure 1–figure supplement 1** [\[1\]](#), **Figure 5** [\[1\]](#), **Figure 8** [\[1\]](#)), chloroplast protrusions (**Figure 5** [\[1\]](#)), stomules (**Figure 7** [\[1\]](#)), or vacuolar RFP intensity (**Figure 8–figure supplement 2** [\[1\]](#)) were acquired with a C2 system (Nikon) equipped with a water-immersion objective lens (CFI Apochromat LWD Lambda S 40XC, NA = 1.15; Nikon). RFP emission was detected at 580–630 nm (bandpass filter RPB580–630; Omega optical) after excitation with a 559.8-nm diode laser; chlorophyll autofluorescence was detected at 660–720 nm following excitation by a 636.5-nm diode laser; the two signals were detected simultaneously. For the simultaneous detection of GFP and chlorophyll, GFP emission was detected at 500–550 nm (bandpass filter RPB500–550; Omega Optical) following excitation by a 489.6-nm diode laser; chlorophyll autofluorescence was detected at 660–720 nm following excitation by a 489.6-nm diode laser.

For the quantification of RCBs and chloroplast protrusions, the second rosette leaves of 21-d-old plants were excised, infiltrated with 10 mM MES-NaOH, pH 5.5, containing 1 μ M concanamycin A (Santa Cruz), and incubated for 1 d in darkness at 23°C. The stock solution was 100 μ M concanamycin A in DMSO. Sucrose (1%, w/v) or full-strength MS salts was added as energy source or nutrients, respectively. The number of accumulated RCBs in a fixed area ($215.04 \times 215.04 \mu\text{m}$ each) was counted. The mean number from four areas of one second rosette leaf per plant was calculated from four or five independent plants. The z-stack images of the 3D region ($215.04 \times 215.04 \times 15 \mu\text{m}$ each) were observed, and the proportion of chloroplasts forming protrusions out of 50 chloroplasts was scored in the region. The protruding structures of chloroplast stroma (approximately 0.5 to 1.5 μm long) containing the chloroplast stroma marker RBCS-mRFP without the chlorophyll signal were defined as chloroplast protrusions in **Figure 5**. The mean from two different regions of one second rosette leaf per plant was calculated from four independent plants.

Chloroplast stromule formation in guard cells was observed in the second rosette leaves of 13-d-old seedlings. The proportion of chloroplasts forming stromules in each pair of guard cells was scored from the z-stack images ($251.04 \times 251.04 \times 20 \mu\text{m}$ each). The mean from 10 stomata of one second rosette leaf per plant was calculated from three individual seedlings. Chloroplast stromule formation in mesophyll cells was observed in the third rosette leaves of 20- or 36-d-old plants. The proportion of chloroplasts forming stromules out of 50 chloroplasts was scored in the z-stack images ($215.04 \times 215.04 \times 15 \mu\text{m}$ each). The mean from two different regions of one third rosette leaf per plant was calculated from four individual plants. Thin, extended tubular structures (less than 1 μm in diameter) containing a chloroplast stroma marker (CT-GFP or RBCS-mRFP) without chlorophyll signals were defined as stromules, as described previously (Hanson and Sattarzadeh, 2011; Brunkard et al., 2015). For the quantification of vacuolar RFP intensity, the second rosette leaves of 21-d-old plants were excised and incubated in 10 mM MES-NaOH, pH 5.5, for 2 d in darkness at 23°C. The RFP intensity in the central area of a mesophyll cell ($15.2 \times 15.2 \mu\text{m}$ each) was measured. The mean from 12 cells of one second rosette leaf per plant was calculated from four individual plants.

Confocal images used to evaluate chloroplast stroma volume (**Figure 1-figure supplement 2**), chloroplast buds, and GFP-ATG8a-labeled structures (**Figure 2-figure supplement 1**; **Figure 6-figure supplement 3** and **4**) were acquired with an LSM900 (Carl Zeiss) or SP8 (Leica) system. To quantify chloroplast stroma volume, the second rosette leaves of 21-d-old plants accumulating RBCS-mRFP were excised at dawn, infiltrated with 10 mM MES-NaOH, pH 5.5, and incubated for 24 h in the dark at 23°C. The z-stack images of the 3D region ($159.73 \times 159.73 \times 30 \mu\text{m}$ each) were acquired, and the volumes of the respective chloroplasts whose entire bodies were captured in the image and could be separated from neighboring chloroplasts as a single structure were measured using Imaris software (Bitplane). Two different regions of one second rosette leaf were observed in five individual plants. Leaves were observed before incubation in the dark as a control. To quantify chloroplast buds, the second rosette leaves of 21- to 23-d-old plants were excised at dawn, infiltrated with 10 mM MES-NaOH, pH 5.5, and incubated for 5 to 9 h in the dark at 23°C. The regions with RBCS-GFP, TOC64-mRFP, KEA1-mRFP, ATPC1-tagRFP, or chlorophyll signals within a chloroplast bud and its neighboring chloroplast or chloroplast envelope were selected as the regions of interest, and the fluorescence intensities per unit area were measured in eight individual plants (**Figure 2-figure supplement 1**). The number of chloroplast buds surrounded by GFP-ATG8a signal in four z-stack images ($159.73 \times 159.73 \times 20 \mu\text{m}$ each) of one leaf per plant was scored from five to six individual plants (**Figure 6-figure supplement 3D** and **3E**).

Protein analysis

The second and third rosette leaves of 21-d-old plants were incubated in darkness in 10 mM MES-NaOH, pH 5.5, for 2 d and then frozen in liquid nitrogen. The leaves from 23-d-old, untreated plants were used as control. The frozen leaves were homogenized using a tissue laser (Qiagen) and a zirconium bead, before being resuspended in homogenization buffer containing 50 mM HEPES-

NaOH, pH 7.5, 16 mM DTT, 10% (v/v) glycerol, and protease inhibitor cocktail (Nacalai). Following centrifugation at 20,630 *g* for 10 min at 4°C, the protein amounts in the supernatants were measured using a 660-nm Protein Assay Reagent (Pierce). The supernatants were then mixed with an equal volume of SDS sample buffer containing 200 mM Tris-HCl, pH 8.5, 20% (v/v) glycerol, 2% (w/v) SDS, and 0.1 M DTT, and incubated for 5 min at 95°C. An equal amount of protein was subjected to SDS-PAGE using TGX FastCast acrylamide gels (Bio-Rad) and transferred to nitrocellulose membrane (Trans-blot turbo transfer pack; Bio-Rad). An anti-RFP 1G9 clone antibody (1:2000, M204-3; MBL) and an anti-cFBPase antibody (1:5000, AS04043; Agrisera) were used as primary antibodies. Goat anti-mouse IgG (H+L) secondary antibody DyLight 800 4X PEG (1:10,000, SA5-35521; Invitrogen) or anti-rabbit HRP secondary antibody (1:10,000, NA934; Cytiba) was used for RFP or cFBPase detection, respectively. The chemiluminescence signals developed with SuperSignal West Dura Extended Duration Substrate (Pierce) and the DyLight 800 fluorescent signals were detected by a ChemiDoc MP system (Bio-Rad). The image processing and the quantification of band intensity were performed using Image Lab Software (Bio-Rad).

H₂O₂ measurements

The amount of H₂O₂ in leaf lysates was measured as previously described ([Chakraborty et al., 2016](#)) with an Amplex Red Hydrogen Peroxide/Peroxidase Assay Kit (Invitrogen) and an Infinite 200 PRO plate reader (Tecan).

Statistical analysis

Statistical analysis in this study was performed with JMP14.3.0 software (SAS Institute). Student's *t*-test or Tukey's test was used to compare paired samples or multiple samples, respectively.

Author contributions

M.I. conceived the study. M.I. performed biochemical assays. M.I., K.O. and T.N. performed two-photon excitation microscopy observations. M.I., S.N., H.I., J.H., and S.H. performed confocal microscopy studies. M.I. wrote the manuscript with the support of all authors.

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Figure Legends

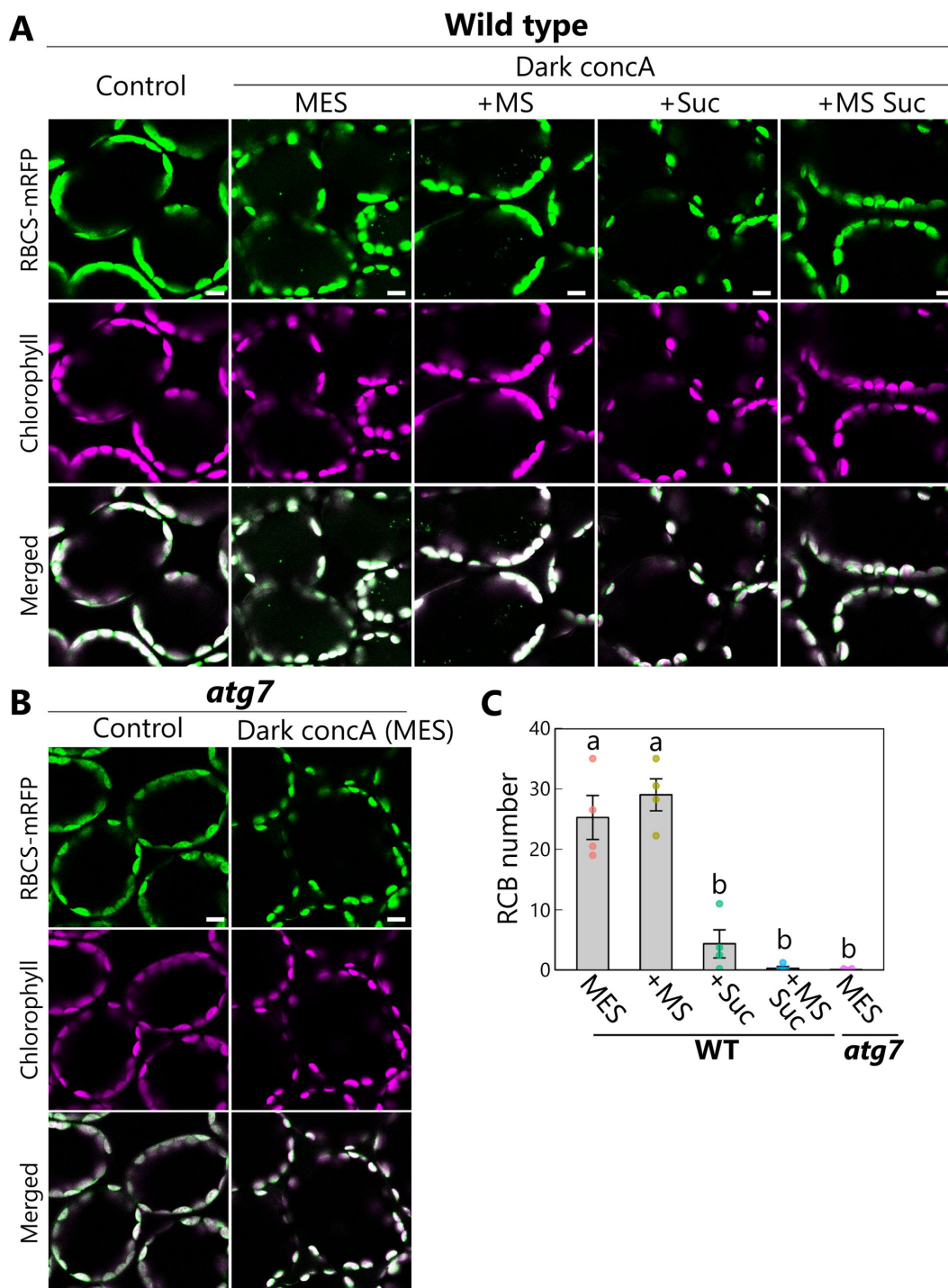


Figure 1–figure supplement 1.

Accumulation of chloroplast stroma components in the vacuole via autophagy.

Confocal images of mesophyll cells from wild-type (A) or *atg7* (B) plants accumulating the chloroplast stroma marker RBCS-mRFP. Second rosette leaves of 21-d-old plants were excised and incubated in 10 mM MES-NaOH containing 1 μ M concanamycin A (conCA) in darkness for 1 d. Sucrose (Suc) or Murashige and Skoog (MS) salts were added as an energy or nutrient source, respectively. Second rosette leaves from nontreated plants are shown as control. Green, RBCS-mRFP; magenta, chlorophyll fluorescence. Scale bars, 10 μ m. The small puncta containing RBCS-mRFP without chlorophyll signal appear green and are Rubisco-containing bodies (RCBs) in the vacuole. (C) Number of accumulated RCBs from the observations described in (A) and (B). Different lowercase letters denote significant differences based on Tukey's test ($P < 0.05$). Values are means $\pm$ SE ($n = 4$). Dots represent individual data points.

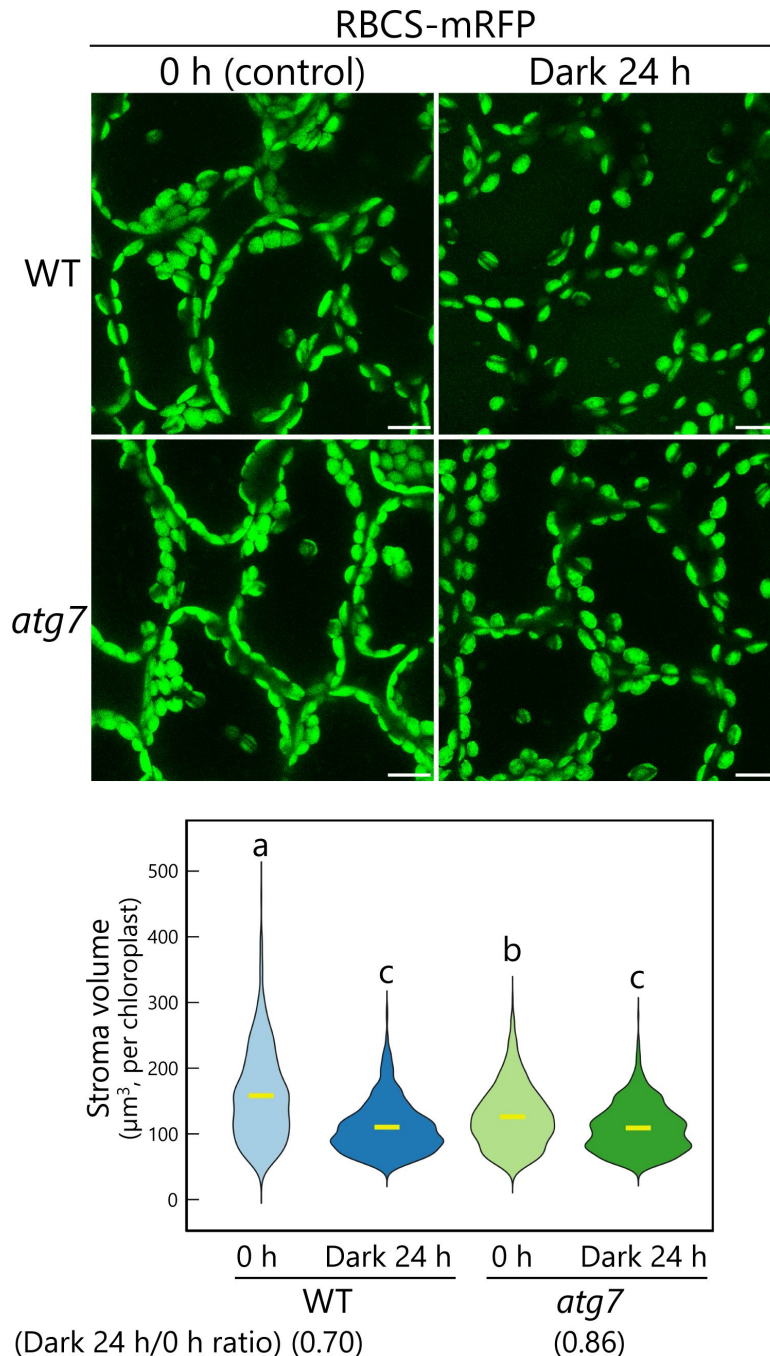


Figure 1-figure supplement 2.

The release of chloroplast buds by autophagy contributes to the decline in chloroplast stroma volume.

(A) Orthogonal projections produced from z-stack images (30 μm in depth) of mesophyll cells from WT or *atg7* leaves accumulating the chloroplast stroma marker RBCS-mRFP. Second rosette leaves of 21-d-old plants were excised at dawn and incubated in 10 mM MES-NaOH in the dark for 24 h. Mesophyll cells were observed before (0 h) and after (Dark 24 h) the treatment. Green, RBCS-mRFP. Scale bars, 10 μm . (B) Violin plot showing the volume of chloroplast stroma labeled by RBCS-mRFP, measured from the observations described in (A). Different lowercase letters denote significant differences based on Tukey's test ($P < 0.05$). Yellow bars indicate mean values. Five individual plants were observed, and the volumes of 564 (WT, 0 h), 554 (WT, dark 24 h), 597 (*atg7*, 0 h), and 725 (*atg7*, dark 24 h) chloroplasts were scored.

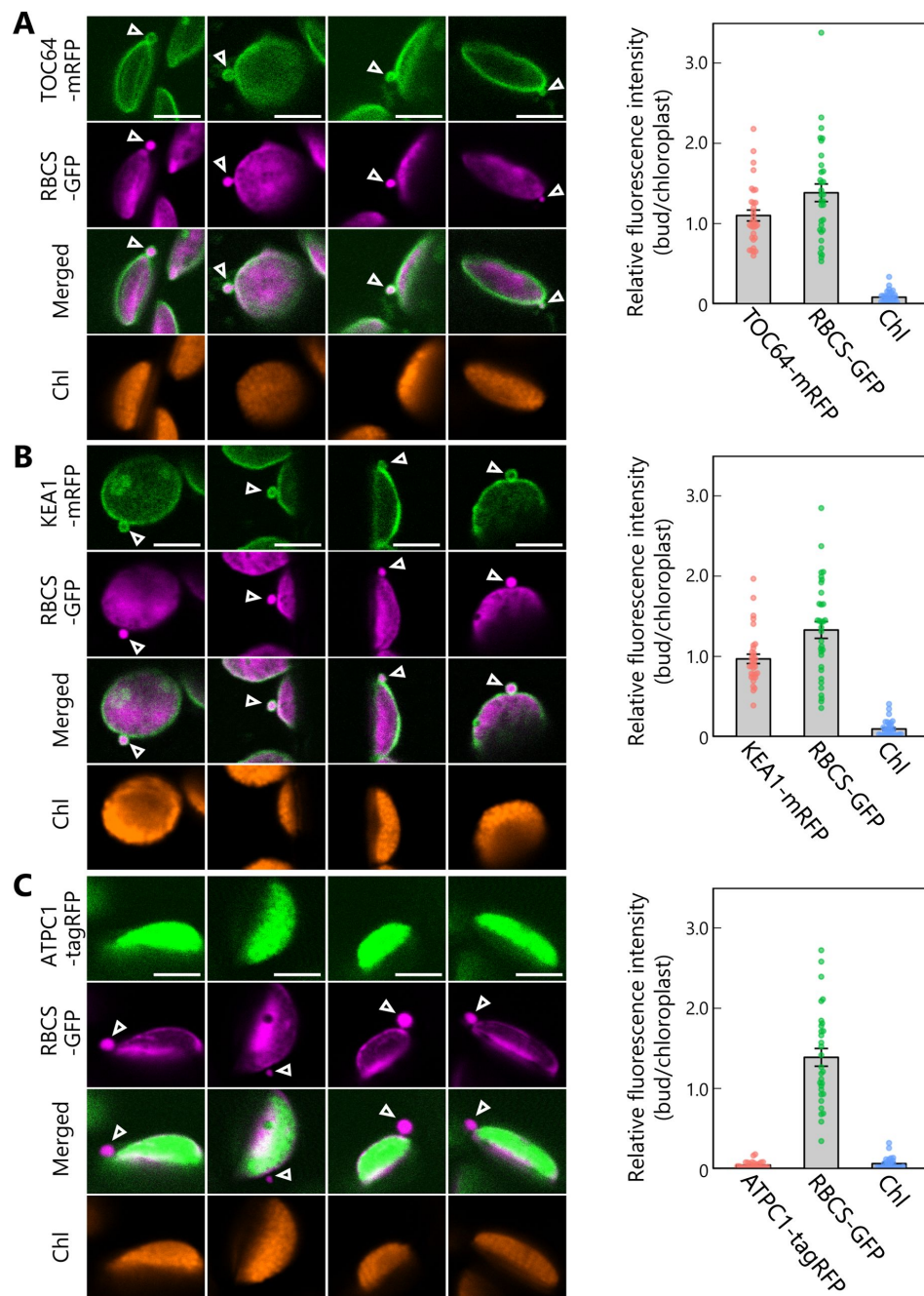


Figure 2-figure supplement 1.

Chloroplast buds contain stroma and envelope proteins.

Second rosette leaves accumulating stromal RBCS-GFP along with outer envelope membrane-bound TOC64-mRFP (A), inner envelope membrane-bound KEA1-mRFP (B), or thylakoid membrane-bound ATPC1-tagRFP (C) were incubated in sugar-free solution in the dark for 5 to 9 h from dawn, and chloroplast budding structures were observed. Second rosette leaves from 21-to 22-d-old plants were used. Arrowheads indicate chloroplast budding structures. Four representative images of chloroplast buds are shown for each plant line. Scale bars, 5 μ m. Green, TOC64-mRFP, KEA1-mRFP, or ATPC1-tagRFP; magenta, RBCS-GFP; orange, chlorophyll (Chl) fluorescence. The graphs show the relative ratio of GFP, RFP, or chlorophyll fluorescence intensity per unit area within chloroplast budding structures vs. their neighboring chloroplasts (bud/chloroplast). The intensities from 33 (A), 32 (B), or 30 (C) chloroplast buds from 8 individual plants were measured. Values are means $\pm$ SE ($n = 30$ -33). Dots represent individual data points in each graph.

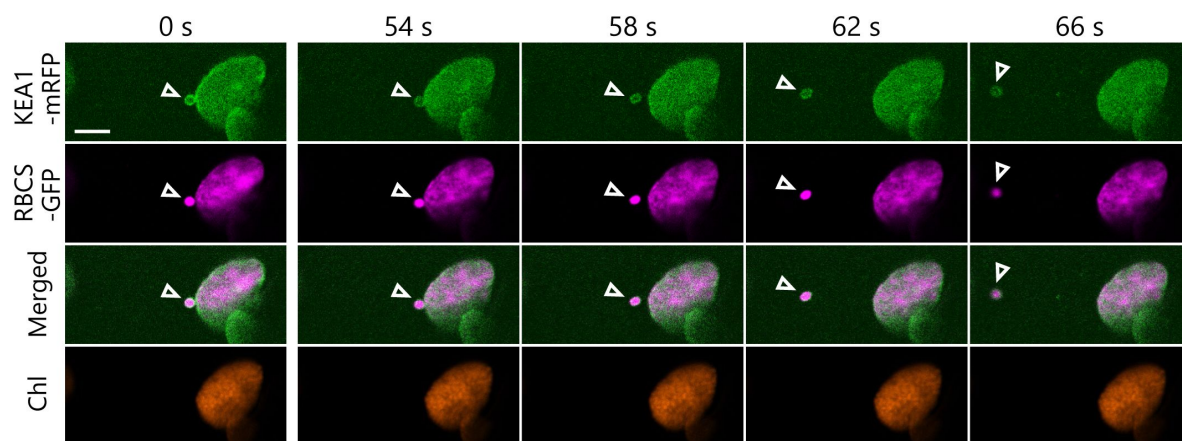


Figure 2-figure supplement 2.

A released chloroplast bud contains an inner envelope marker protein.

Time-lapse observations of Arabidopsis mesophyll cells accumulating the chloroplast stroma marker RBCS-GFP along with inner envelope membrane-bound KEA1-mRFP. The second rosette leaf from a 22-d-old plant was incubated in sugar-free solution in the dark for 6 h from dawn and observed. The images are still frames from Video 5. Time scales above the images indicate the elapsed time from the start of the video. Arrowheads indicate chloroplast budding structures. Scale bars, 5 μ m. Green, KEA1-mRFP; magenta, RBCS-GFP; orange, chlorophyll (Chl) fluorescence.

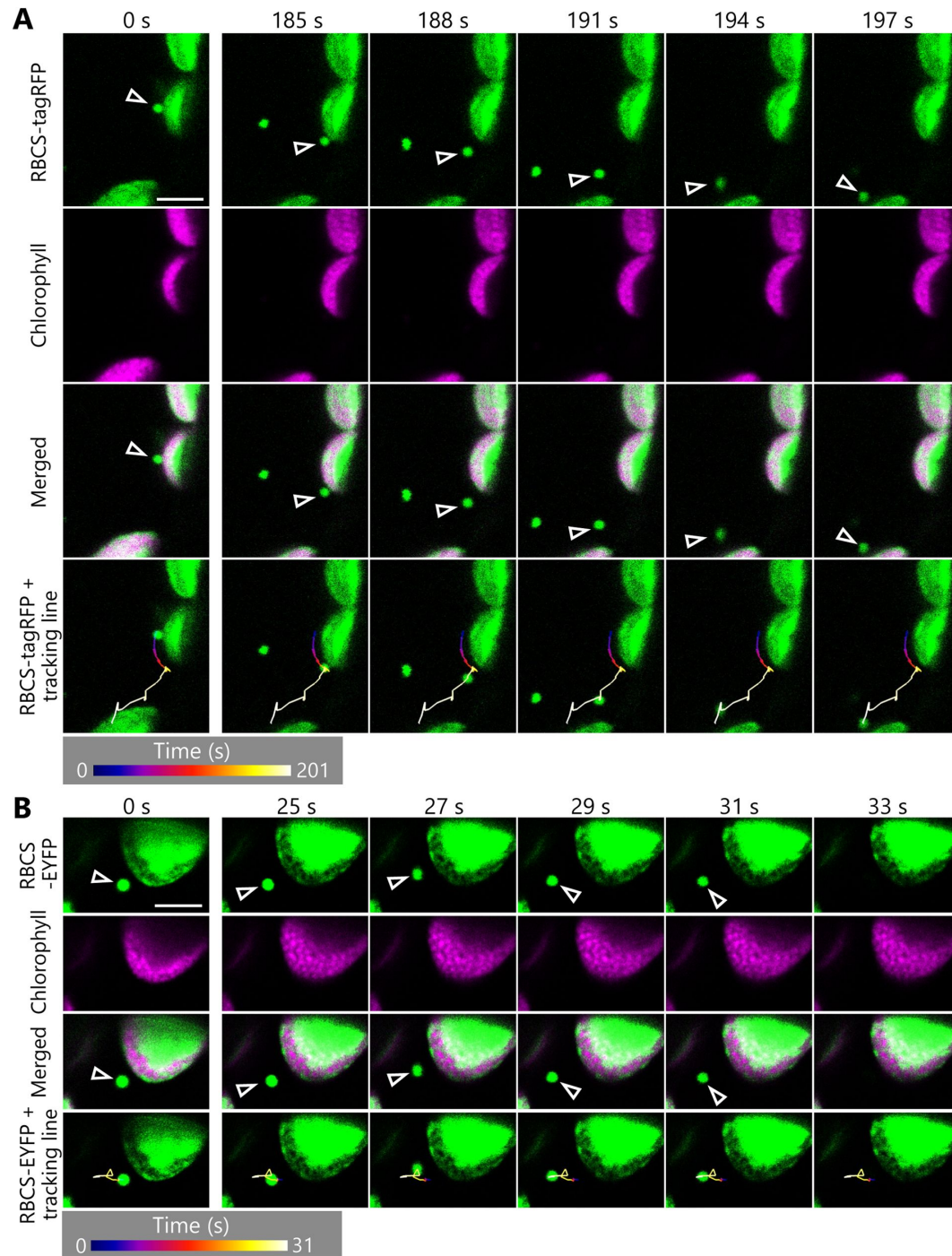


Figure 3-figure supplement 1.

Rubisco-containing bodies appear to begin random movement during their tracking.

A leaf accumulating RBCS-tagRFP (A) or RBCS-EYFP (B) was incubated in sugar-free solution in the dark for 7 to 8 h from dawn, and the transport of a Rubisco-containing body was tracked. Second rosette leaves from 22-d-old plants were used. Arrowheads indicate a tracked RCB. The images are still frames from Videos 8 (A) or 9 (B), respectively. The images show the time when RCBs appeared to begin random movement (185–197 s in A or 25–33 s in B). Time scales above the images indicate the elapsed time from the start of the respective videos. Green, RBCS-tagRFP or RBCS-EYFP; magenta, chlorophyll fluorescence. Scale bars, 5 μ m. The color of the track line changes over time, as indicated by the color bars below the images.

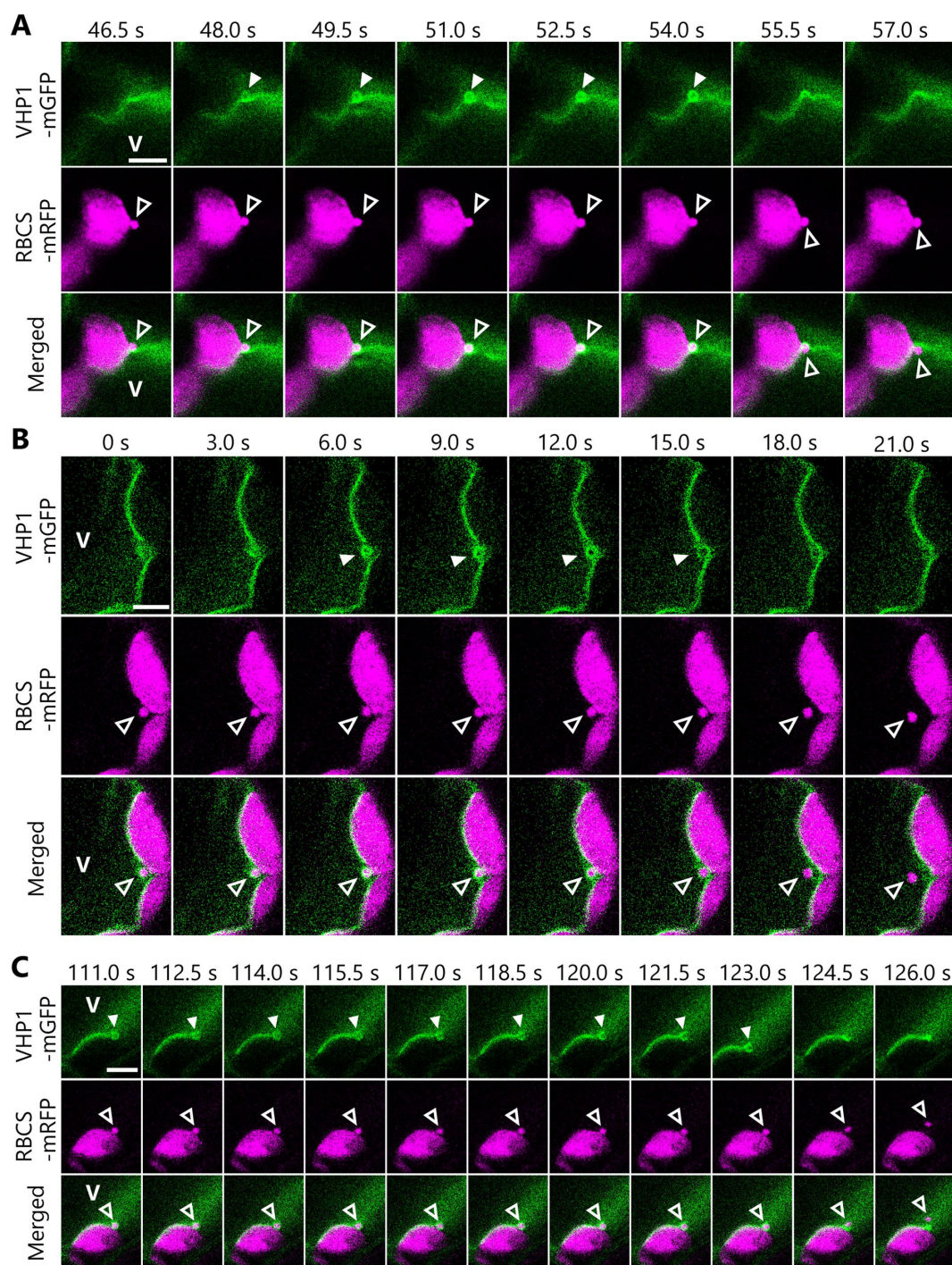


Figure 4-figure supplement 1.

Additional observations of the incorporation of Rubisco-containing bodies into the vacuolar lumen.

Additional time-lapse data of the experiments described in [Figure 4](#). The second rosette leaves from 21-(A) or 24-d-old (B, C) plants were used. The images in (A), (B), and (C) are still frames from Videos 12, 13, and 14, respectively. Time scales above the images indicate the elapsed time from the start of the respective videos. Open arrowheads indicate an RCB engulfed by the vacuolar membrane. Closed arrowheads indicate the vacuolar membrane structure that engulfs an RCB. V indicates the region of the vacuolar lumen. Green, VHP1-mGFP; magenta, RBCS-mRFP. Scale bars, 5 μ m.

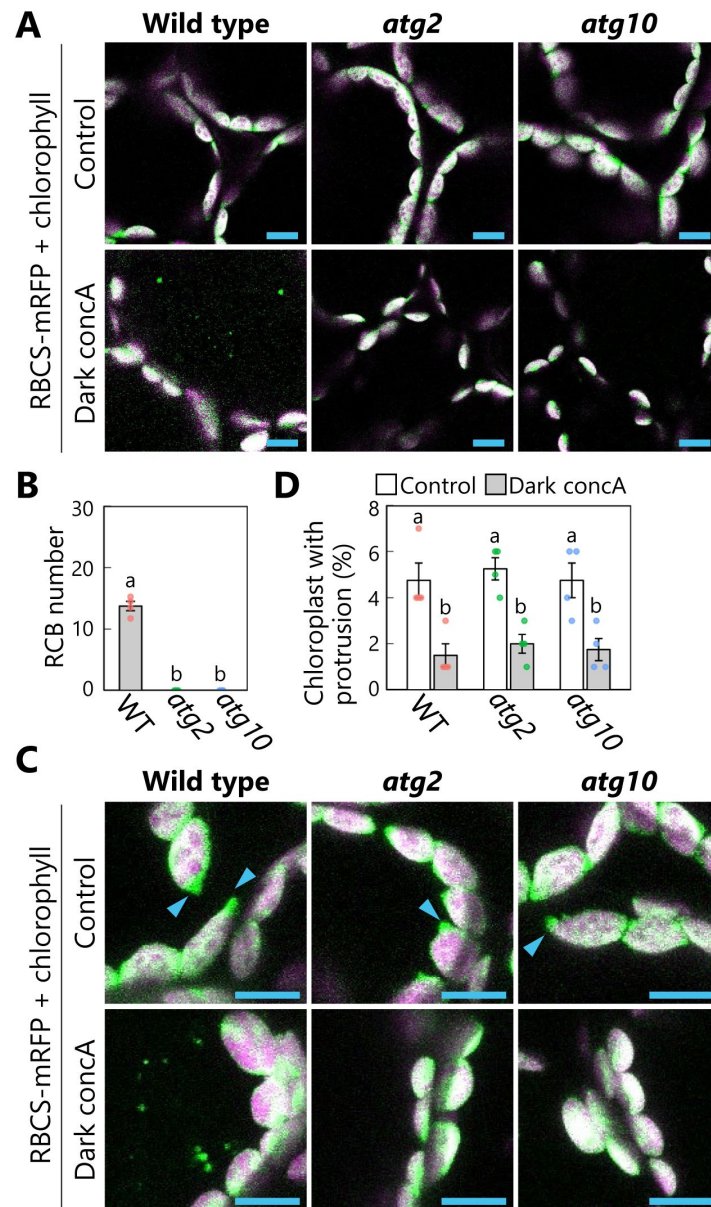


Figure 5-figure supplement 1.

Chloroplast protrusions do not increase in *atg2* or *atg10* mutant leaves during a 1-d dark treatment.

The experiments described in **Figure 5** were performed on leaves from wild-type (WT), *atg2*, or *atg10* plants accumulating the chloroplast stroma marker RBCS-mRFP. Scale bars, 10 μ m. Green, RBCS-mRFP; magenta, chlorophyll fluorescence. Only the merged channels are shown. The overlapping regions of RBCS-mRFP and chlorophyll signals appear white. Small puncta containing RBCS-mRFP without chlorophyll signal appear green and are RCBs in the vacuole. Arrowheads indicate the structures that were counted as a chloroplast protrusion in (D). Different lowercase letters denote significant differences based on Tukey's test ($P < 0.05$). Values are means $\pm$ SE ($n = 4$). Dots represent individual data points in each graph.

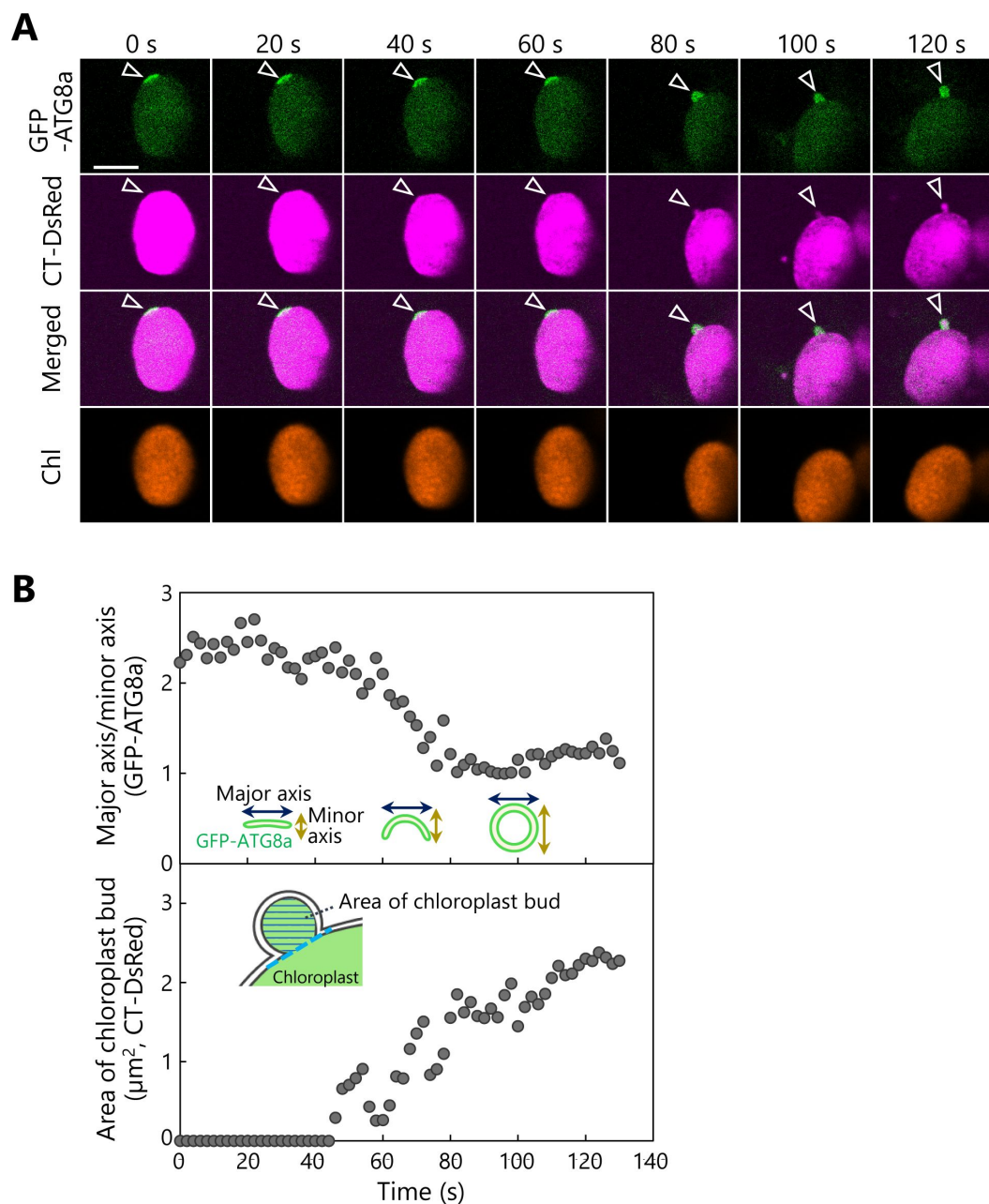


Figure 6—figure supplement 1.

Another observation of the budding of the isolation membrane-associated site in a chloroplast.

A leaf accumulating the chloroplast stroma marker CT-DsRed and the isolation membrane marker GFP-ATG8a was incubated in sugar-free solution in darkness for 6 h from dawn and then observed. A second rosette leaf from a 22-d-old plant was used. Arrowheads indicate the position of the chloroplast-associated isolation membrane. The images are still frames from Video 17. Time scales above the images indicate the elapsed time from the start of the video. Green, GFP-ATG8a; magenta, CT-DsRed. Scale bars, 5 μm . (B) Time-dependent changes in the ratio of the major axis to the minor axis in the GFP-ATG8a-labeled isolation membrane (top) or in the area of the chloroplast budding structure (bottom), as measured from the images in (A).

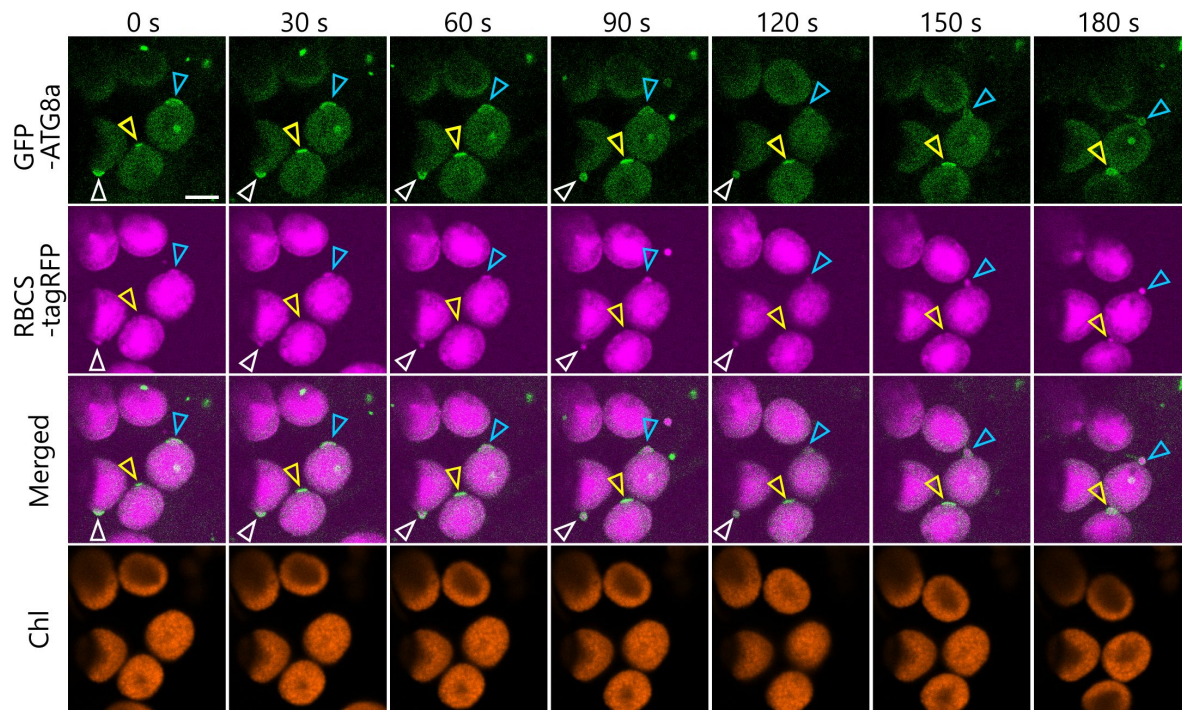


Figure 6-figure supplement 2.

Chloroplast buds surrounded by the isolation membrane appear in multiple chloroplasts.

A leaf accumulating the chloroplast stroma marker RBCS-tagRFP and the isolation membrane marker GFP-ATG8a was incubated in sugar-free solution in darkness for 5 h from dawn and then observed. A second rosette leaf from a 21-d-old plant was used. White, yellow, and blue arrowheads indicate the isolation membrane-associated site in different chloroplasts, respectively. The images are still frames from Video 18. Time scales above the images indicate the elapsed time from the start of the video. Green, GFP-ATG8a; magenta, RBCS-tagRFP; orange, chlorophyll (Chl) fluorescence. Scale bars, 5 μ m.

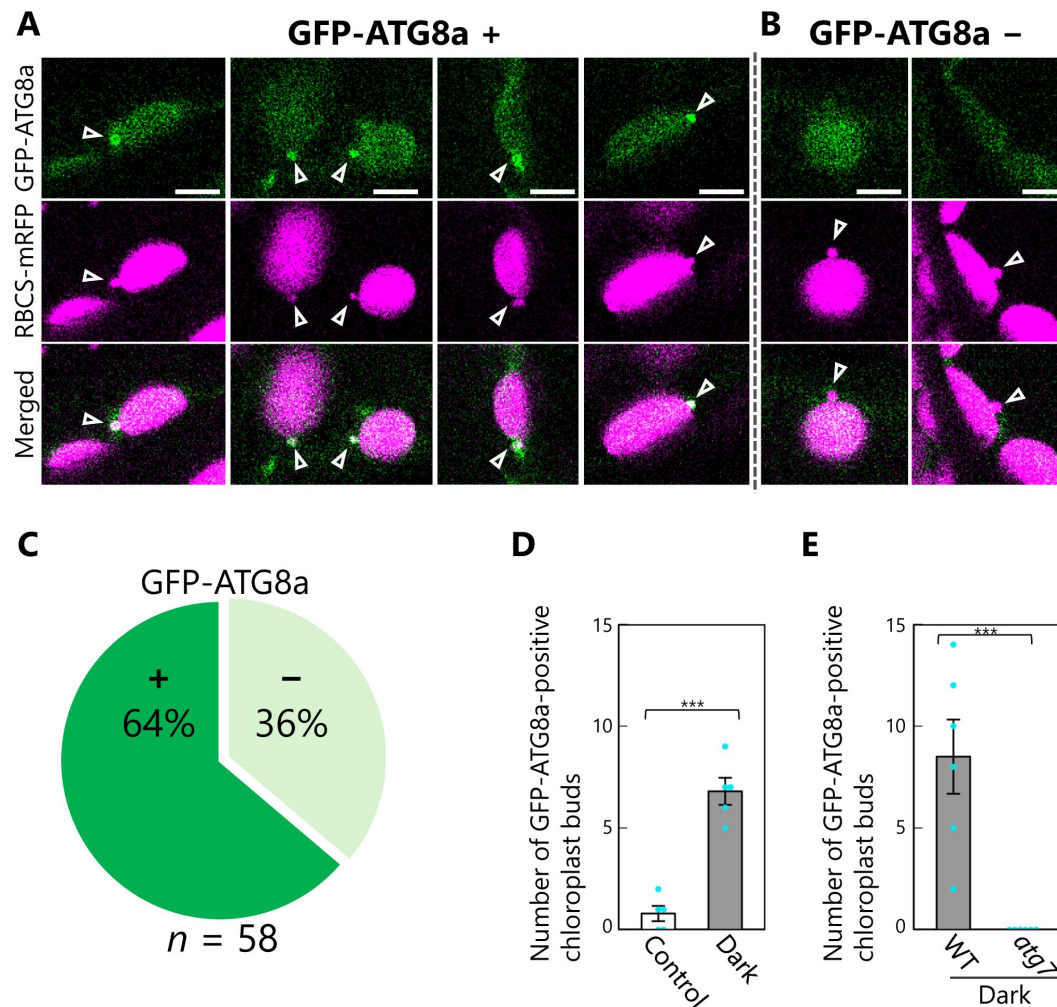


Figure 6-figure supplement 3.

Proportion of chloroplast buds engulfed by GFP-ATG8a-labeled isolation membranes.

Leaves accumulating the chloroplast stroma marker RBCS-mRFP and the isolation membrane marker GFP-ATG8a were incubated in sugar-free solution in the dark for 5 to 9 h from dawn, and chloroplast budding structures were observed. Second rosette leaves from 22-to 23-d-old plants were used. (A and B) Four representative images of chloroplasts buds that are surrounded by GFP-ATG8a signals (A) or two representative images of chloroplast buds that are not labeled by GFP-ATG8a signals (B) are shown. Arrowheads indicate chloroplast budding structures. Green, GFP-ATG8a; magenta, RBCS-mRFP. Scale bars, 5 μ m. (C) The proportions of chloroplast buds that are labeled by GFP-ATG8a signals (+) or not (-) were calculated from observations of 58 chloroplast buds from 8 individual plants. (D) The number of GFP-ATG8a-positive chloroplast buds, as shown in (A), were counted in the fixed regions from leaves incubated in the dark for 5 to 9 h from dawn (Dark) or from untreated leaves of plants grown under normal conditions (Control). Five individual plants were observed. (E) Leaves from wild-type (WT) or *atg7* plants accumulating RBCS-mRFP and GFP-ATG8a were incubated in the dark for 5 to 9 h from dawn, and the number of GFP-ATG8a-positive chloroplast buds was counted. Six individual plants were observed. Values are means $\pm$ SE ($n = 5-6$). Asterisks denote significant differences based on *t*-test (***, $P < 0.001$). Dots represent individual data points in each graph.

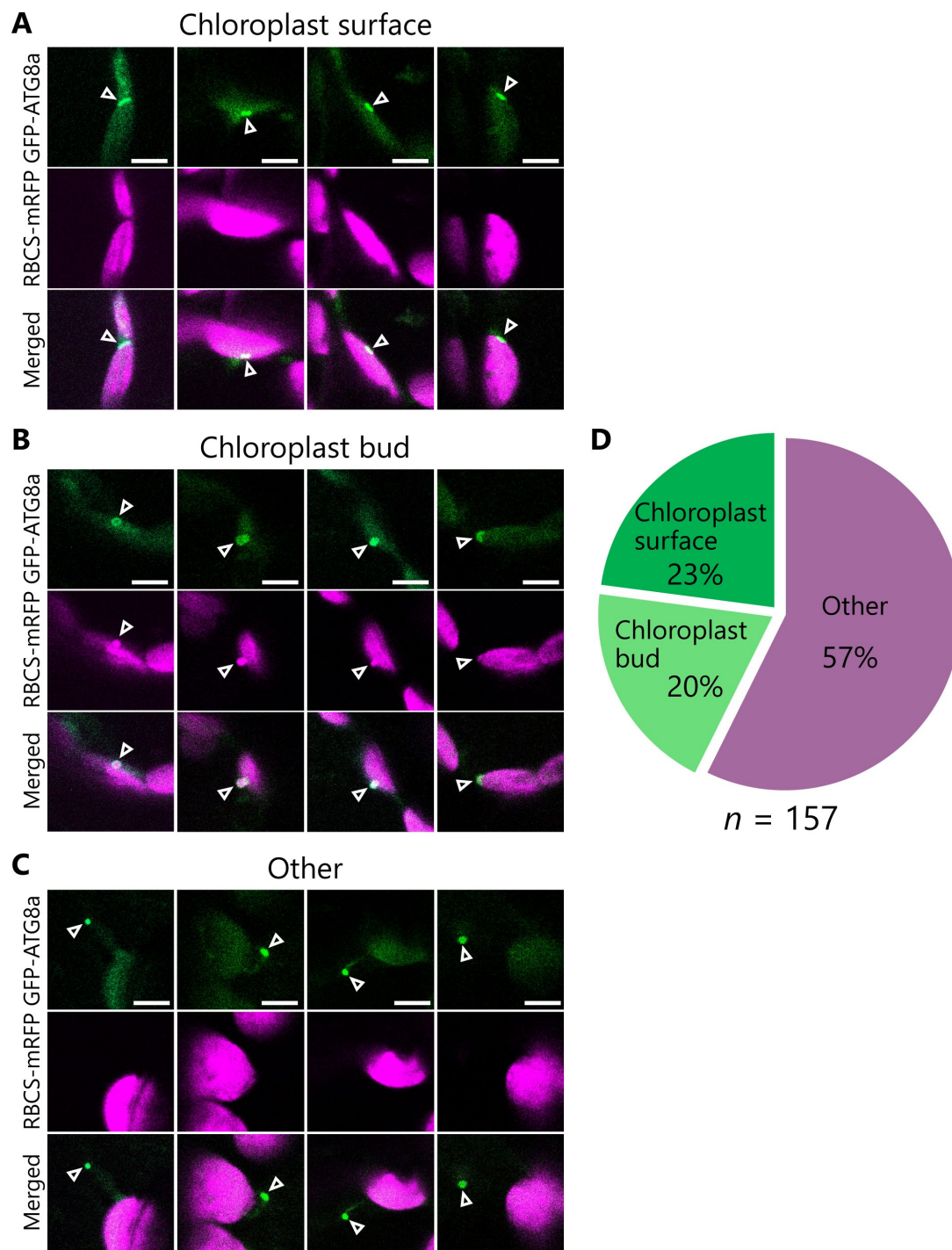


Figure 6—figure supplement 4.

Proportion of GFP-ATG8a-labeled structures that are associated with chloroplasts.

Leaves accumulating the chloroplast stroma marker RBCS-mRFP and the isolation membrane marker GFP-ATG8a were incubated in sugar-free solution in the dark for 5 to 9 h from dawn, and GFP-ATG8a-labeled structures in mesophyll cells were observed. Second rosette leaves from 22-d-old plants were used. (A to C) GFP-ATG8a-positive structures were classified into three types: flattened structures associated with the chloroplast surface (A), spherical or curved structures associated with chloroplast budding structures (B), and other structures (C). Four representative images of each type are shown. Green, GFP-ATG8a; magenta, RBCS-mRFP. Scale bars, 5 μ m. (D) The proportions of the three types of GFP-ATG8a-labeled structures were calculated from observations of 157 GFP-ATG8a-labeled structures from 13 individual plants.

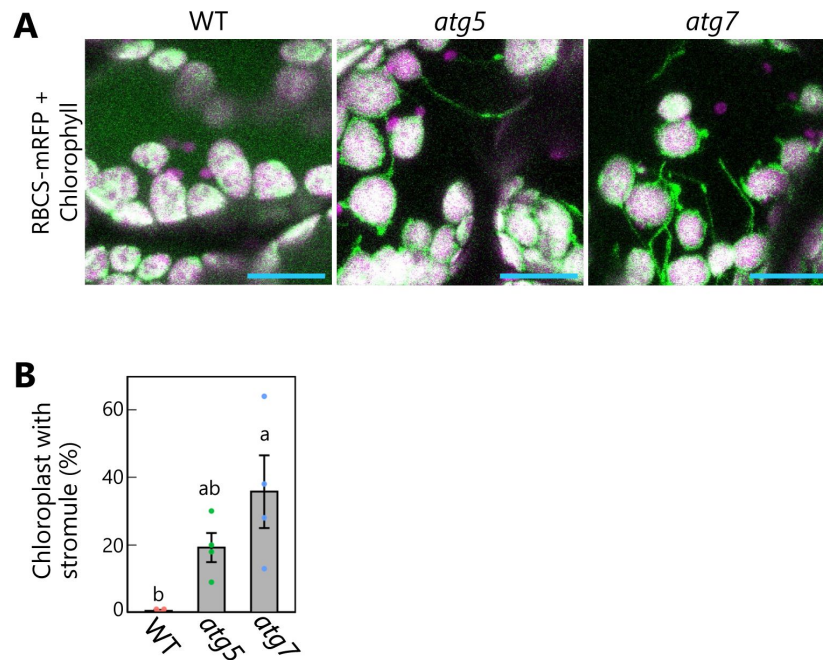


Figure 7-figure supplement 1.

Autophagy deficiency activates stromule formation from mesophyll chloroplasts accumulating RBCS-mRFP in senescing leaves.

(A) Orthogonal projections produced from z-stack images (15 μ m in depth) of mesophyll cells from leaves of WT, *atg5*, and *atg7* plants accumulating RBCS-mRFP. Third rosette leaves from 36-d-old plants were observed. Green, RBCS-mRFP; magenta, chlorophyll fluorescence. Only the merged channels are shown. Spherical bodies only exhibiting chlorophyll signal represent a portion of chloroplasts in pavement cells. Scale bars, 20 μ m. (B) Percentage of chloroplasts forming stromules out of 50 chloroplasts in mesophyll cells, from the observations described in (A). Different lowercase letters denote significant differences based on Tukey's test ($P < 0.05$). Values are means $\pm$ SE ($n = 4$). Dots represent individual data points in the graph.

Figure 7-figure supplement 2.

Autophagy deficiency does not activate stromule formation from mesophyll chloroplasts in young leaves.

Orthogonal projections produced from z-stack images (15 μm in depth) of mesophyll cells from WT, *atg5*, *atg7*, *sid2*, *sid2 atg5*, and *sid2 atg7* leaves accumulating CT-GFP. Third rosette leaves from 20-d-old plants were observed. Green, CT-GFP; magenta, chlorophyll fluorescence. Only the merged channels are shown. Scale bars, 20 μm .

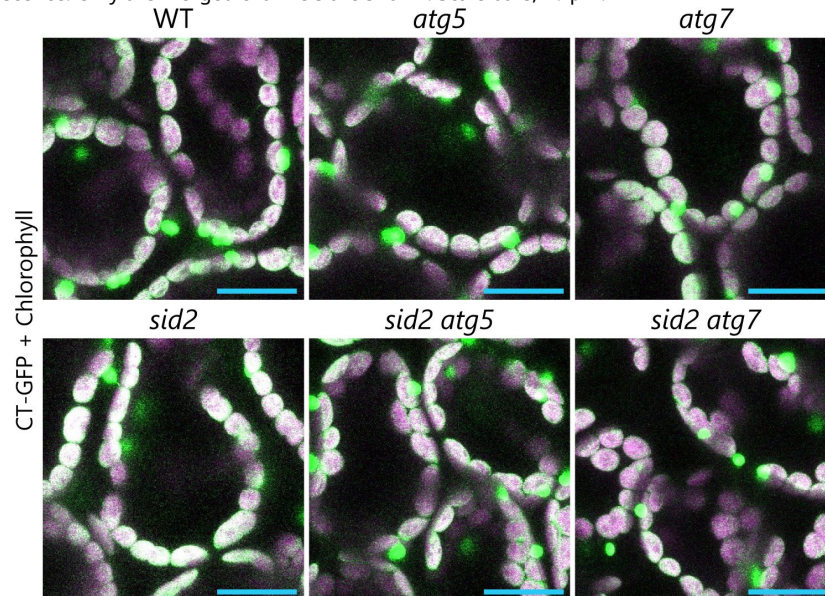
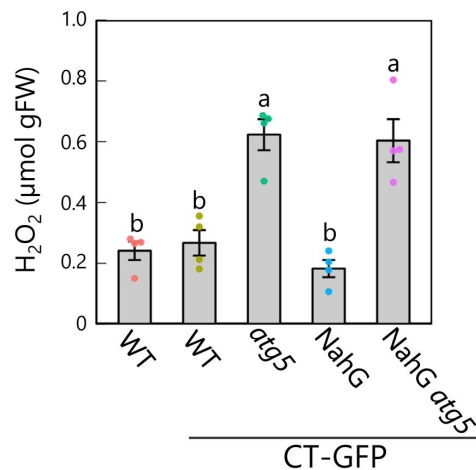


Figure 7-figure supplement 3.

NahG expression does not counteract the increase in hydrogen peroxide content produced by the *atg5* mutation.

Hydrogen peroxide (H_2O_2) contents in third rosette leaves from 36-d-old plants. WT plants without any transgenic construct and CT-GFP-expressing plants in the WT, *atg5*, NahG, and *atg5* NahG backgrounds were used. Different lowercase letters denote significant differences based on Tukey's test ($P < 0.05$). Values are means $\pm$ SE ($n = 4$). Dots represent individual data points.



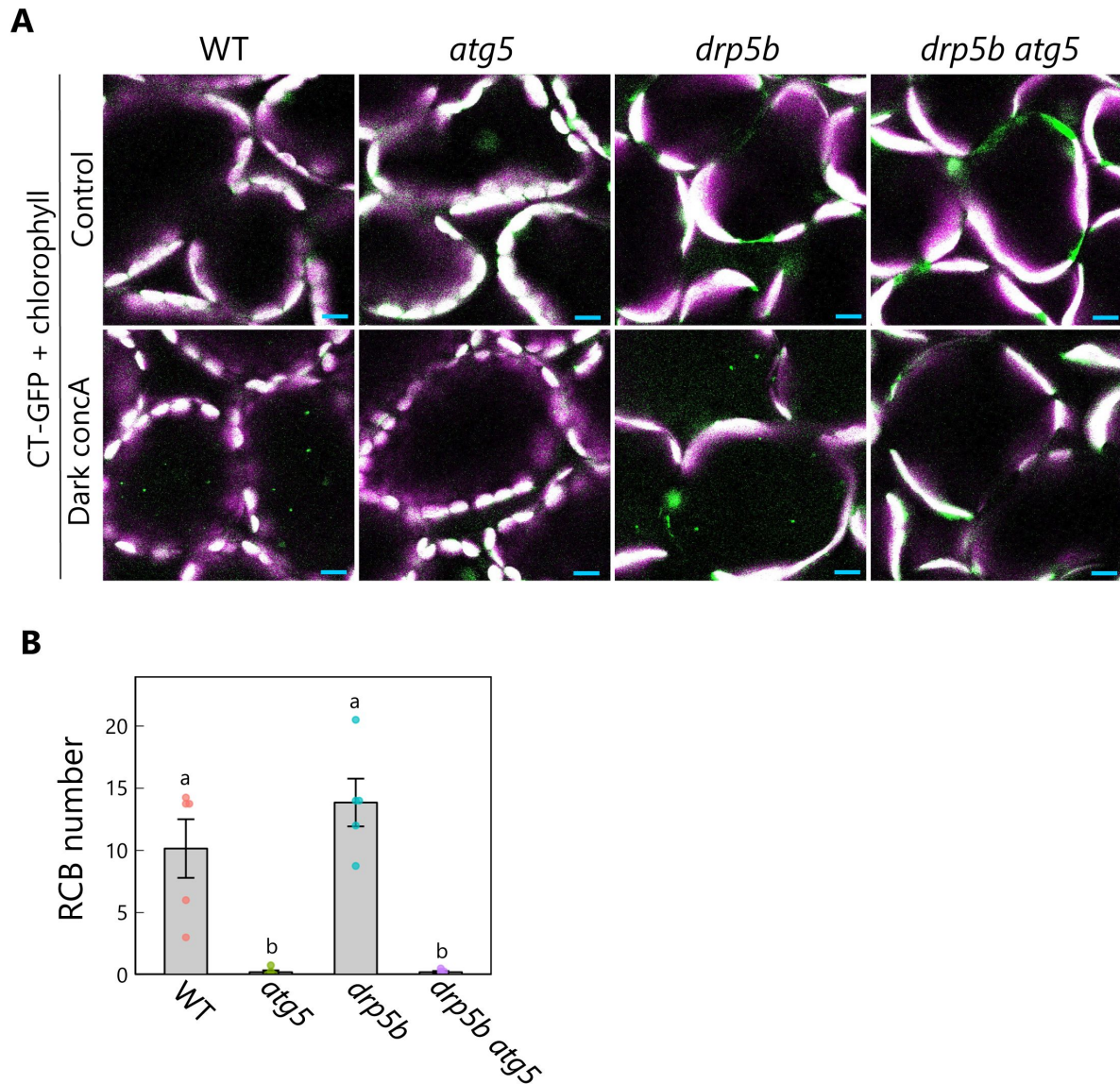


Figure 8–figure supplement 1.

Production of Rubisco-containing bodies in *drp5b* mutants is ATG5 dependent.

(A) Confocal images of mesophyll cells from wild-type (WT), *atg5*, *drp5b*, and *drp5b atg5* leaves accumulating the stroma marker CT-GFP. Second rosette leaves from 21-d-old plants were incubated in sugar-free solution containing 1 μ M concanamycin A (conCA) in darkness for 1 d. Second rosette leaves from untreated plants were used as control. Green, CT-GFP; magenta, chlorophyll fluorescence. Only the merged channels are shown. Small puncta containing CT-GFP without chlorophyll signal appear as green and are RCBs in the vacuole. Scale bars, 10 μ m. (B) Number of accumulated RCBs, counted from the observations of leaves incubated in the dark with concA described in (A). Different lowercase letters denote significant differences based on Tukey's test ($P < 0.05$). Values are means $\pm$ SE ($n = 5$). Dots represent individual data points.

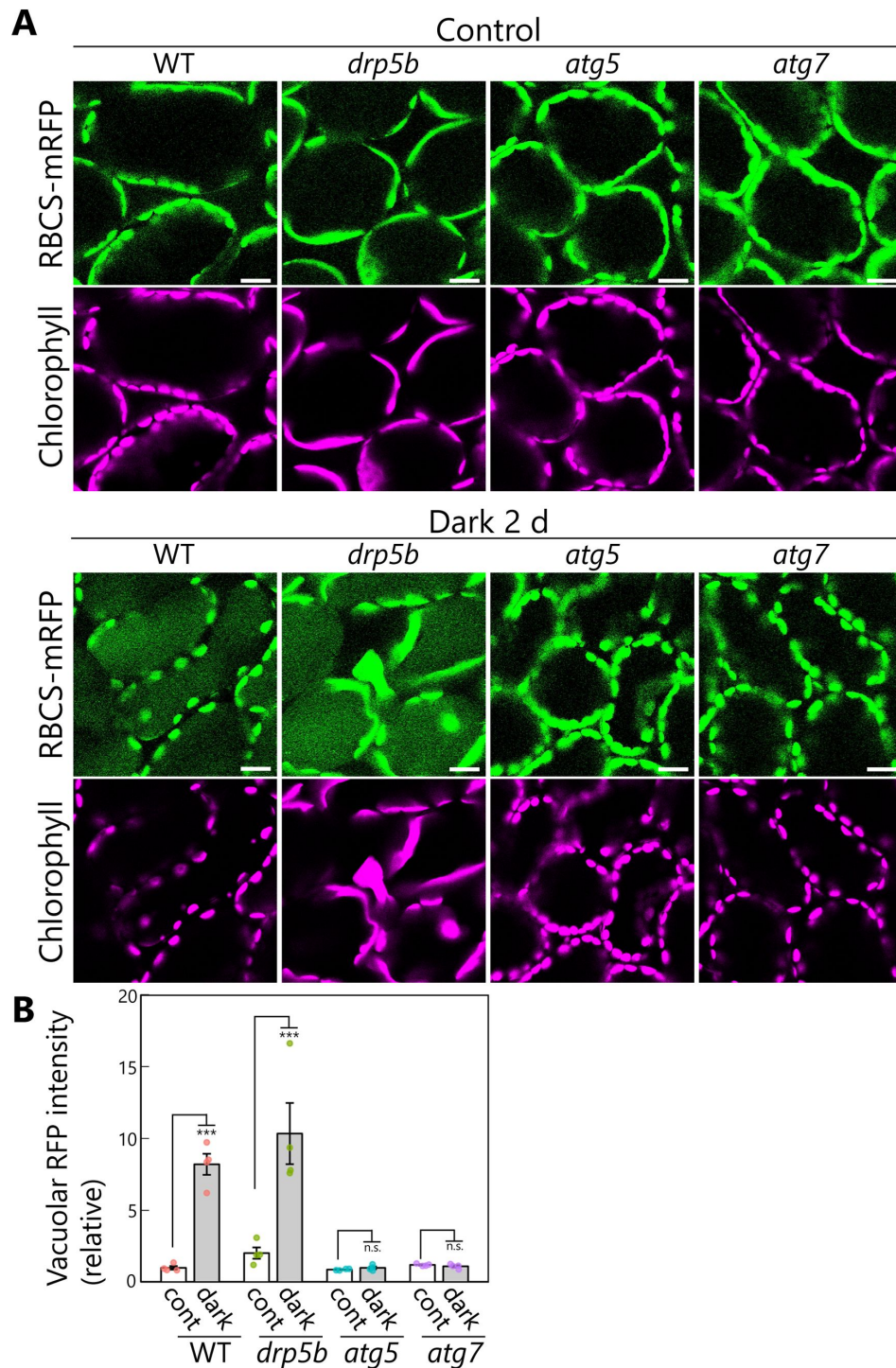


Figure 8-figure supplement 2.

Vacuolar accumulation of stromal marker proteins in sugar-starved leaves.

(A) Confocal images of mesophyll cells from wild-type (WT), *drp5b*, *atg5*, and *atg7* leaves accumulating the stroma marker RBCS-mRFP. Second rosette leaves from 21-d-old plants were incubated in sugar-free solution in darkness for 2 d. Second rosette leaves from untreated plants are shown as control. Green, RBCS-mRFP; magenta, chlorophyll fluorescence. Scale bars, 20 μ m. (B) RFP intensity in the vacuolar lumen, as (B) measured from the observations described in (A) and shown relative to that of untreated WT plants, which was set to 1. Asterisks denote significant differences based on *t*-test (***, $P < 0.001$; n.s., not significant). Values are means $\pm$ SE ($n = 4$). Dots represent individual data points.

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Reviewer #1 (Public review):

Summary:

The authors demonstrated that carbon depletion triggers the autophagy-dependent formation of Rubisco Containing Bodies, which contain chloroplast stroma material, but exclude thylakoids. The authors show that RCBs bud directly from the main body of chloroplasts rather than from stromules and that their formation is not dependent on the chloroplast fission factor DRP5. The authors also observed a transient engulfment of the RCBs by the tonoplast during delivery to the vacuolar lumen.

Strengths:

The authors demonstrate that autophagy-related protein 8 (ATG8) co-localizes to the chloroplast demarking the place for RCB budding. The authors provide good-quality time-lapse images and co-localization of the markers corroborating previous observations that RCBs contain only stroma material and do not include thylakoid. The text is very well written and easy to follow.

Weaknesses:

The study adds more valuable descriptive information about the previously published phenomenon of RCB formation under carbon starvation but does not reveal the putative mechanisms governing formation of RCBs and their release to the vacuole.

Comments on revised version:

The authors have done an impressive job revising the manuscript and addressed my comments. The authors clarified previous ambiguities and the new version of the manuscript greatly benefits from the provided quantifications and adjusted discussion.

<https://doi.org/10.7554/eLife.93232.2.sa3>

Reviewer #2 (Public review):

This manuscript proposed a new link between the formation of chloroplast budding vesicles (Rubisco-containing bodies [RCBs]) and the development of chloroplast-associated autophagosomes. The authors' previous work demonstrated two types of autophagy pathways involved in chloroplast degradation, including piecemeal degradation of partial chloroplast and whole chloroplast degradation. However, the mechanisms underlying piecemeal degradation are largely unknown, particularly regarding the initiation and release of the budding structures. Here, the authors investigated the progression of piecemeal-type chloroplast trafficking by visualizing it with a high-resolution time-lapse microscope. They provide evidence that autophagosome formation is required for the initiation of chloroplast budding, and that stromule formation is not correlated with this process. In addition, the authors also demonstrated that the release of chloroplast-associated autophagosome is independent of a chloroplast division factor, DRP5b.

Overall, the findings are interesting, and in general, the experiments are very well executed.

Comments on revised version:

The authors have generally addressed all of my concerns (and the other reviewer's) and adapted the manuscript where necessary. The revised version has significantly improved the manuscript. From my perspective there are no further concerns.

Reviewer #3 (Public review):

Summary:

Regulated chloroplast breakdown allows plants to modulate these energy-producing organelles, for example during leaf aging, or during changing light conditions. This manuscript investigates how chloroplasts are broken down during light-limiting conditions.

The authors present very nice time lapse imaging of multiple proteins as buds form on the surface of chloroplasts and pinch away, then associate with the vacuole. They use mutant analysis and autophagy markers to demonstrate that this process requires the ATG machinery, but not dynamin-related proteins that are required for chloroplast division. The manuscript concludes with discussion of an internally-consistent model that summarizes the results.

Strengths:

The main strength of the manuscript is the high-quality microscopy data. The authors use multiple markers and high-resolution timelapse imaging to track chloroplast dynamics under light limiting conditions.

Weaknesses:

The main weakness of the manuscript is the limited quantitative data. While it can be challenging to quantify dynamic intracellular events, quantification of these processes is important to appreciate the significance of these findings.

<https://doi.org/10.7554/eLife.93232.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public Review):

Summary:

The authors demonstrated that carbon depletion triggers the autophagy-dependent formation of Rubisco Containing Bodies, which contain chloroplast stroma material, but exclude thylakoids. The authors show that RCBs bud directly from the main body of chloroplasts rather than from stromules and that their formation is not dependent on the chloroplast fission factor DRP5. The authors also observed a transient engulfment of the RCBs by the tonoplast during delivery to the vacuolar lumen.

Strengths:

The authors demonstrate that autophagy-related protein 8 (ATG8) co-localizes to the chloroplast demarking the place for RCB budding. The authors provide good-quality time-lapse images and co-localization of the markers corroborating previous observations that RCBs contain only stroma material and do not include thylakoid. The text is very well written and easy to follow.

Weaknesses:

A significant portion of the results presented in the study comes across as a corroboration of the previous findings made under different stress conditions: autophagy-dependent formation of RCBs was reported by Ishida et al in 2009. Furthermore, some included results are not of particular relevance to the study's aim. For example, it is unclear what is the importance of the role of SA in the formation of stromules, which do not serve as an origin for the RCBs. Similarly, the significance of the transient engulfment of RCBs by the tonoplast remained elusive. Although it is indeed a curious observation, previously reported for peroxisomes, its presentation should include an adequate discussion maybe suggesting the involved mechanism. Finally, some conclusions are not fully supported by the data: the suggested timing of events poorly aligns between and even within experiments mostly due to high variation and low number of replicates. Most importantly, the discussion does not place the findings of this study into the context of current knowledge on chlorophagy and does not propose the significance of the piece-meal vs complete organelle sequestration into the vacuole under used conditions, and does not dwell on the early localization of ATG8 to the future budding place on the chloroplast.

We performed additional experiments with biological replicates that involved quantification. The results of these experiments validate the findings of this study. We also revised the Discussion section, which now includes a discussion of the interplay between piecemeal-type and entire-organelle-type chloroplast autophagy and the relevance of autophagy adaptor and receptor proteins to the localization of ATG8 on the chloroplast surface. Accordingly, the first subheading section in the Discussion became too long. Therefore, we divided it into two subheading sections. We believe that the revisions successfully address the weaknesses pointed out by the reviewer and enhance the importance of the current study. Below is a detailed description of the improvements made to our manuscript in response to the reviewer comments.

Reviewer #1 (Recommendations For The Authors):

It would be great if the authors kindly used numbered lines to facilitate the review process.

We have added line numbers to the text of the revised version of the manuscript.

The authors use the words "budding", "protrusion" and "stromule formation" interchangeably in some parts of the text. For the sake of clarity, it would be best to be consistent in the terminology and possibly elaborate on the exact differences between these structure types and the criteria by which they were identified.

We have checked all of the text and improved the consistency of the terminology. An important finding of this study is that chloroplasts form budding structures at the site associated with ATG8. These structures then divide to become a type of autophagic cargo termed a Rubisco-containing body. We therefore mainly use the terms "bud" and "budding" throughout the text. In the experiments shown in Figure 5, we considered the possibility that chloroplast protrusions accumulate in leaves of *atg* mutants and do not divide because the mutants cannot create autophagosomes. Therefore, the word "protrusion" was used to describe the results shown in Figure 5 in which the proportion of chloroplasts forming protrusions was scored. In the revised text, the word "protrusion" is only used in descriptions of Figure 5. Previous reports define stromules as thin, tubular, extended structures (less than 1 μ m in diameter) of the plastid stroma (Hanson and Sattarzadeh, 2011; Brunkard et al., 2015). In the revised text, the word "stromules" is used to describe the structures defined in these previous reports. We have added definitions of each term to the Introduction, Methods

and Results sections where appropriate (lines 57–58, 160–162, 247–249, 313–316, 655–658, 668–670).

Pages 3-4: the authors observed budding of the chloroplasts within a few minutes - it would be helpful to specify that time was probably counted from the first observation of budding, not from the start of the dark treatment, and also specify the exact treatment duration for each of the experiments.

The time scales in the figures do not represent the time from the start of the dark treatment. Instead, they describe the duration from the start of the time-lapse videos that were used to generate the still images. Therefore, the indicated time scales are almost the same as the duration from the start of the observations of each target structure (chloroplast buds or GFPATG8a-labeled structures). As described in the Methods section, leaves were incubated in darkness for 5 to 24 h to induce sugar starvation. Such sugar-starved leaves were subjected to live-cell monitoring for the target structures. Since Arabidopsis leaves accumulate starch as a stored sugar source (Smith and Stitt, 2007; Usadel et al., 2008), dark treatment lasting several minutes is not sufficient for the starch to be consumed and sugar starvation to be induced. To avoid confusion, we have added definitions of the time scales to the legends of figures containing the results of time-lapse imaging. We have also specified the durations of dark treatments used to obtain the respective results in the legends.

Figure 6: the time scale for complete autophagosome formation is in the range of 100-120 sec, how do these results align with the results shown in Figures 3B and C, where complete autophagosomes are suggested to be released into the vacuole after 73.8 sec. Furthermore, another structure is suggested to be formed within 50 sec. Such experiments possibly require a large number of replicates to estimate representative timing.

As mentioned in the previous response, the time scales in still frames represent the duration from the start of the corresponding video. Leaves incubated in darkness for 5 to 24 h were subjected to live-cell imaging. When we identified the target structures, e.g., GFP-ATG8a-labeled structures on the surfaces of chloroplasts (Figure 6) or chloroplast budding structures (Figure 3), we began to track these structures. Therefore, the time scales in the figures do not align to a common time axis. We revised the descriptions about Figure 3 and Figure 6 in the Results section to clearly explain that the time points in each experiment merely indicates the time of one observation.

The authors might want to consider using arrows to indicate structures of interest in all movies and figures.

We have added arrows to indicate the structures of interest in the starting frames of all videos. We hesitate to add arrows to highlight RCBs accumulating in the vacuole (Figure 1-figure supplement 1, Figure 5 and Figure 8) and stromules (Figure 7) because many arrows would be required, which would obscure large portions of the images. We believe that the images without arrows clearly represent the appearance of RCBs or stromules and that their quantification (Figure 1-figure supplement 1C, Figure 5B, Figure 5-figure supplement 1B, Figure 7B, 7D, 7F, and Figure 8B) well supports the results.

Figure 7 Supplement 1: do the authors detect complete chloroplasts in the vacuole of atg7 and sid2/atg7?

We did not observe the vacuolar transport of whole chloroplasts in *atg7* or *atg7 sid2* plants under our experimental conditions. The figure below (Figure 1 for Response to reviewers) shows images of mesophyll cells from a leaf (third rosette leaf of a 20-d-old plant) of *atg7*

accumulating chloroplast stroma-targeted GFP (CT-GFP); this is from the previous version of Figure 7-figure supplement 1. Indeed, some GFP bodies exhibiting strong stromal GFP (CTGFP) signals appeared in the central area of the cell (arrowheads in A). However, such bodies were chloroplasts in epidermal cells. The 3D images (B) and cross-section image (x to z axis) of the region highlighted by the blue dotted line (C) indicate that such GFP bodies are the edges of chloroplasts that localize on the abaxial side of the observed region. Because CT-GFP expression was driven by the 35S promoter, strong GFP signals appeared in chloroplasts in epidermal cells in addition to chloroplasts in mesophyll cells. Previous studies using the same transgenic lines also showed that chloroplasts in epidermal cells exhibit strong GFP signals (Kohler et al., 1997; Caplan et al., 2015; Lee et al., 2023). RBCS2B promoter do not label the chloroplasts in epidermal cells (new Figure 7-figure supplement 1). Additionally, because the borders between the mesophyll cell layer and the epidermal cell layer are not even, chloroplasts in epidermal cells are sometimes visible during observations of mesophyll cells. Such detection more frequently occurs during the acquisition of z-stack images. This point was more precisely demonstrated in our previous study with the aid of Calcofluor white staining of cell walls (Nakamura et al., 2018). Please see Supplemental Figure S3 in our previous report. To avoid any misunderstanding, we replaced the image of the leaf from *atg7* in the revised figure, which is now Figure 7-figure supplement 2, with an image of another region to more precisely visualize mesophyll cells in this plant line.

Author response image 1.

Mesophyll cells in a leaf of *atg7* accumulating stromal CT-GFP, reconstructed from the data shown in the previous version of Figure 7-figure supplement 1. (A) Individual channel images (CT-GFP and chlorophyll) from the merged orthogonal projection image shown in the previous version of Figure 7-figure supplement 1. The right panel shows the enhanced chlorophyll signal to clearly visualize the chloroplasts in epidermal cells. Green, CTGFP; magenta, chlorophyll fluorescence. Scale bar, 20 μ m. (B) 3D structure of the merged image shown in (A). (C) Images of the cross section indicated by the blue dotted line (a to b) in B. Arrowheads indicate the edges of chloroplasts in epidermal cells.

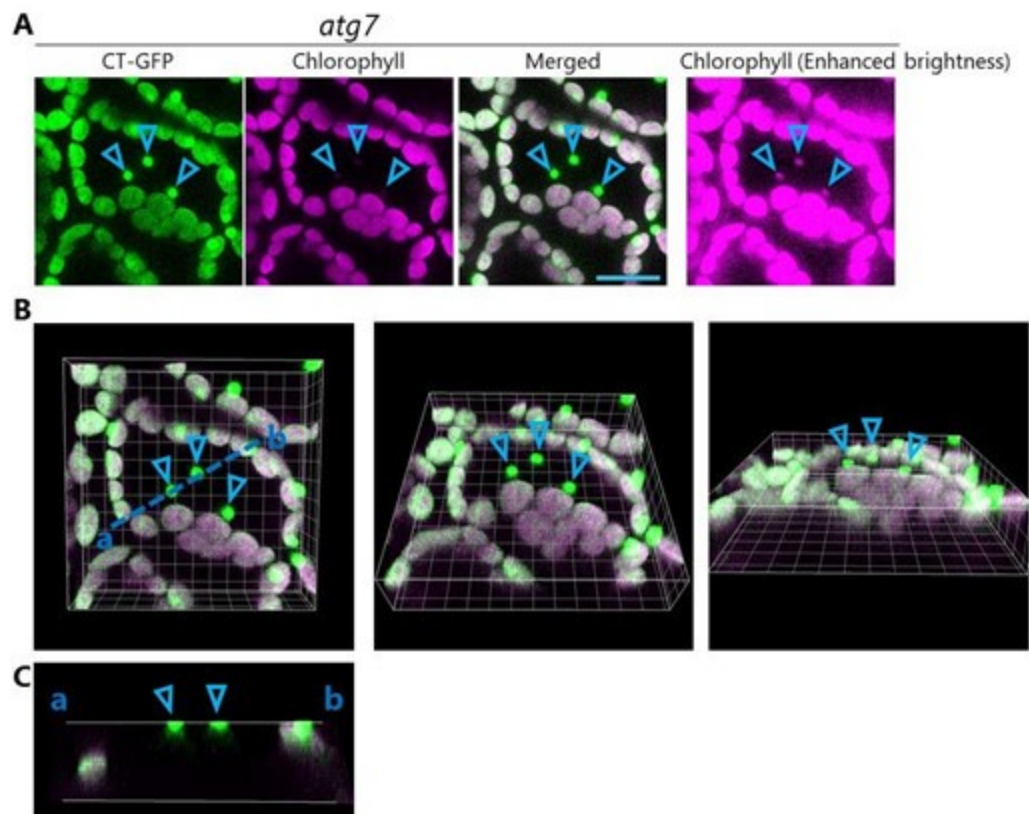


Figure 8: it would be interesting to hear the authors' opinion on why they observed a significant increase in RCBs number in the *drp5b* mutant background

We have added a discussion of this issue to the revised manuscript (lines 445–459). We now have two hypotheses to explain this issue. One hypothesis is that the impaired chloroplast division due to the *drp5b* mutation reduces energy availability and thus activates chloroplast autophagy. The other hypothesis is that the *drp5b* mutation impairs the type of chlorophagy that degrades whole chloroplasts, and thus piecemeal-type chloroplast autophagy via Rubisco-containing bodies is activated. However, we do not have any experimental evidence supporting either hypothesis.

Reviewer #2 (Public Review):

This manuscript proposed a new link between the formation of chloroplast budding vesicles (Rubisco-containing bodies [RCBs]) and the development of chloroplast-associated autophagosomes. The authors' previous work demonstrated two types of autophagy pathways involved in chloroplast degradation, including piecemeal degradation of partial chloroplast and whole chloroplast degradation. However, the mechanisms underlying piecemeal degradation are largely unknown, particularly regarding the initiation and release of the budding structures. Here, the authors investigated the progression of piecemeal-type chloroplast trafficking by visualizing it with a high-resolution time-lapse microscope. They provide evidence that autophagosome formation is required for the initiation of chloroplast budding, and that stromule formation is not correlated with this process. In addition, the authors also demonstrated that the release of chloroplast-associated autophagosome is independent of a chloroplast division factor, DRP5b.

Overall, the findings are interesting, and in general, the experiments are very well executed. Although the mechanism of how Rubisco-containing bodies are processed is still unclear, this study suggests that a novel chloroplast division machinery exists to facilitate chloroplast autophagy, which will be valuable to investigate in the future.

Reviewer #2 (Recommendations For The Authors):

Below are some specific comments.

(1) In Supplement Figure 1B, there is no chloroplast stromule in RBCS-mRFP x atg7-2 plants under dark treatment with ConA, but in Figure 7A, there are stromules in CT-GFP x atg7-2 plants. How to explain such a discrepancy? Did the authors check the chloroplast morphology of RBCS-mRFP x atg7-2 plants in different developmental stages? Will it behave the same as CT-GFP x atg7-2 under the same condition as in Figure 7A?

As described in the text, the ages and conditions of the leaves shown in Figure 1–figure supplement 1 and Figure 7 are different. In Figure 1–figure supplement 1, second rosette leaves from 21-d-old plants were incubated in the dark with concanamycin A for 1 d. In Figure 7E and 7F, we explored the condition under which mesophyll chloroplasts in *atg* leaves actively form stromules to assess how a deficiency in autophagy is related to stromule formation. We found that late senescing leaves (third rosette leaves from 36-d-old plants) of *atg5* and *atg7* plants accumulated many stromules without additional treatment (Figure 7). It is not surprising that the chloroplast morphologies shown in Figures 1 and 7 are different because the leaf ages and conditions are largely different.

However, we agree that the differences in chloroplast stroma–targeted GFP and RBCS-mRFP might influence the visualization of stromules. For instance, fluorescent protein–labeled RBCS proteins are incorporated into the Rubisco holoenzyme, comprising eight RBCS and eight RBCL proteins (Ishida et al., 2008; Ono et al., 2013). Such a large protein complex might not accumulate in stromules. Therefore, we examined the chloroplast morphology in late senescing leaves (third rosette leaves from 36-d-old plants) from WT, *atg5*, and *atg7* plants harboring *ProRBCS:RBCS-mRFP*, as you suggested. Mesophyll chloroplasts formed many stromules in *atg5* and *atg7* leaves but not in WT leaves (Figure 7–figure supplement 1). These results indicate that RBCS-mRFP can be used to visualize stromules and that the differences in chloroplast morphology between Figure 1–figure supplement 1 and Figure 7 cannot be attributed to the different marker proteins used. A previous study also indicated that Rubisco is present in plastid stromules (Kwok and Hanson, 2004).

(2) In Figure 2, the author showed that the outer envelope marker Toc64 was colocalized with chloroplast buds. How about proteins in the inner envelope membrane of chloroplasts?

We generated Arabidopsis plants expressing red fluorescent protein–tagged K⁺ EFFLUX ANTIporter 1 (KEA1), a chloroplast inner envelope membrane protein (Kunz et al., 2014; Boelter et al., 2020). We found that the chloroplast buds visualized by RBCS-GFP were also marked by KEA1-mRFP (Figure 2–figure supplement 1B). We observed the transport of such buds (Figure 2–figure supplement 2). These results strengthen our claim that autophagy degrades chloroplast stroma and envelope components as a type of specific cargo termed a Rubisco-containing body. The descriptions about this additional experiment are in lines 181–187.

(3) In Figure 3, how many RCBs were tracked for the trafficking analysis to raise the conclusion that the vesicle was released into the vacuole around 73.8s?

We apologize for our confusing explanation in the previous version of the manuscript. The time point “73.8 s” merely indicates the time of one observation, as shown in Figure 3. This time does not represent the common timing of vacuolar release of a Rubisco-containing body. As we explained in the response to the comments from reviewer 1, we subjected leaves that were incubated in the dark for several hours to live-cell imaging assays to observe chloroplast morphology in sugar-starved leaves. The time scales of each still frame represent the time from the start of the corresponding video. Therefore, the time points in the respective figures do not align to a common time axis, and the number “73.8 s” is not important. We attempted to emphasize that the type of movement of Rubisco-containing bodies changes during their tracking shown in Figure 3. Based on this finding, we hypothesized that the Rubisco-containing bodies are released into the vacuolar lumen when they initiate random movement. Therefore, we expected that the interaction between the Rubisco-containing bodies and the vacuolar membrane could be captured, and we therefore turned our attention to the dynamics of the vacuolar membrane in subsequent experiments. Accordingly, our observations of the vacuolar membrane allowed us to visualize the release of the Rubisco-containing body into the vacuole (Figure 4). We rephrased these sentences (lines 212–219) to avoid confusion and to explain this idea accurately. We also performed tracking experiments of Rubisco-containing bodies to strengthen the finding that the type of movement of the bodies changes during tracking (Figure 3-figure supplement 1, Videos 8 and 9).

(4) I do believe the conclusion that vacuolar membranes incorporate RCBs into the vacuole in Figure 4. However, it will be more convincing if images of higher quality are provided.

We tried to acquire images that more clearly show the morphology of the vacuolar membrane during the incorporation of the Rubisco-containing body. We obtained the images in Figure 4A using a standard type of confocal microscope, the LSM 800 (Carl Zeiss), and obtained the images in Figure 4B using the Airyscan Fast acquisition mode, a hyper-resolution microscope mode, in the LSM 880 system (Carl Zeiss). We performed additional experiments with another type of confocal microscope, the SP8 (Leica; Figure 4-figure supplement 1A to 1C, Videos 12–14). The quality of the images from these experiments was as high as possible under the experimental conditions (equipment and plant materials). In general, increasing the image resolution during time-lapse imaging with a confocal microscope requires reducing the time resolution. However, the transport of a Rubisco-containing body occurs relatively quickly: Its engulfment by the vacuolar membrane takes place for just a few seconds (Figure 4, Figure 4-figure supplement 1). We could therefore not reduce the time resolution further to better capture the morphology of the vacuolar membrane.

(5) In Figure 7G, the authors concluded that SA and ROS might be the cause of the extensive formation of stromules. How about the H₂O₂ level in NahG and atg5 NahG plants? Compared with sid2, NahG appeared to completely inhibit stromule formation in atg5. Will this be related to ROS levels?

We measured the hydrogen peroxide (H₂O₂) contents in NahG *atg5* plants and *atg5* single mutant plants and found that their leaves accumulate more H₂O₂ than those of wild-type or NahG plants (Figure 7-figure supplement 3). Since we have only maintained fresh seeds of NahG *atg5* plants harboring the 35S promoter-driven chloroplast stroma-targeted GFP (*Pro35S:CT-GFP*) construct, we first confirmed that CT-GFP accumulation does not affect the measurement of H₂O₂ content. H₂O₂ levels were similar between wild-type leaves and CT-GFP-expressing leaves. A comparison among *Pro35S:CT-GFP* expressing lines in the wild-type, *atg5*, NahG, and NahG *atg5* backgrounds revealed enhanced accumulation of H₂O₂ in the

atg5 and NahG *atg5* genotypes compared with the wild-type and NahG genotypes. This finding is consistent with the results of histological staining of H₂O₂ using 3,3'-diaminobenzidine (DAB) in a previous study (Yoshimoto et al., 2009).

It is unclear why NahG expression inhibited stromule formation more strongly than the *sid2* mutation in the *atg5* mutant background, as you pointed out (Figure 7A–D). NahG catabolizes salicylic acid (SA), whereas *sid2* mutants are knockout mutants of *ISOCHORISMATE SYNTHASE1* (*ICS1*), a gene required for SA biosynthesis. Plants have two metabolic routes for SA biosynthesis: The isochorismate synthase (ICS) pathway and the phenylalanine ammonia-lyase (PAL) pathway. Furthermore, Arabidopsis plants contain two ICS homologs: ICS1 and ICS2. Previous studies have revealed that ICS1 (*SID2*) is the main player for SA biosynthesis in response to pathogen infection (Delaney et al., 1994). Another study revealed drastically lower SA contents in the leaves of both *sid2* single mutants and NahG-expressing plants compared with those of wild-type plants (Abreu and Munné-Bosch, 2009). Therefore, it is clear that the *sid2* single mutation sufficiently inhibits SA accumulation in Arabidopsis leaves. However, low levels of SA biosynthesis through ICS1-independent routes might influence stromule formation in leaves of *sid2 atg5* and *sid2 atg7*. Because a previous study demonstrated that the *sid2* single mutation sufficiently suppresses the SA hyperaccumulation-related phenotypes of *atg* plants (Yoshimoto et al., 2009), we believe that the use of the *sid2* mutation was adequate to assess the effects of SA on stromule formation that actively occurs in the *atg* plants examined in this study.

(6) In Supplement Figure 7, I have noticed that there are still some CT-GFP signals (green dots) in the vacuoles of the *atg7* mutant, are they RCBs? If so, how can this phenomenon be explained?

As we explained in the response to the comment from Reviewer 1, CT-GFP-labeled bodies are chloroplasts in the epidermal cell layer. Please see our response to Reviewer 1's comment about Figure 7 and the associated figure (Figure 1 for Response to reviewers). The CT-GFP-labeled dots (arrowheads) are the edges of chloroplasts and localize on the abaxial side of the observed region. The dots have faint chlorophyll signals. This phenomenon is much more clear in the image with enhanced brightness (right panel in A). Since the bodies are merely the edges of epidermal chloroplasts, their chlorophyll signals are faint. Therefore, these bodies are not Rubisco-containing bodies but are instead simply the edges of chloroplasts in the epidermal cell layer.

(7) On page 24, the second paragraph, lines 12–14, the authors claim that no receptors similar to those involved in mitophagy that bind to LC3 (*ATG8*) have been established in chloroplasts. Actually, it has been reported that a homologue of mitophagy receptor, *NBR1*, acts as an autophagy receptor to regulate chloroplast protein degradation (Lee et al, 2023, *Elife*; Wan et al, 2023, *EMBO Journal*). Although I do think *NBR1* is not involved in RCBs based on these reports, these findings should be discussed here.

Thank you for this good suggestion. We have added a discussion about this important point to the Discussion section, along with the relevant citations (lines 482–502).

(8) In the figure legend, the details of the experiments will be better provided, such as leaves stages (Figure 1, Figure 5...), the number of chloroplasts analyzed (Figure 7...). This can help the readers to follow.

Thank you for highlighting this. We have checked all of the figure legends and added descriptions of the leaf stages and experimental conditions.

Reviewer #3 (Public Review):

Summary:

Regulated chloroplast breakdown allows plants to modulate these energy-producing organelles, for example during leaf aging, or during changing light conditions. This manuscript investigates how chloroplasts are broken down during light-limiting conditions.

The authors present very nice time-lapse imaging of multiple proteins as buds form on the surface of chloroplasts and pinch away, then associate with the vacuole. They use mutant analysis and autophagy markers to demonstrate that this process requires the ATG machinery, but not dynamin-related proteins that are required for chloroplast division. The manuscript concludes with a discussion of an internally-consistent model that summarizes the results.

Strengths:

The main strength of the manuscript is the high-quality microscopy data. The authors use multiple markers and high-resolution time-lapse imaging to track chloroplast dynamics under light-limiting conditions.

Weaknesses:

The main weakness of the manuscript is the lack of quantitative data. Quantification of multiple events is required to support the authors' claims, for example, claims about which parts of the plastid bud, about the dynamics of the events, about the colocalization between ATG8 and the plastid stroma buds, and the dynamics of this association. Without understanding how often these events occur and how frequently events follow the manner observed by the authors (in the 1 or 2 examples presented in each figure) it is difficult to appreciate the significance of these findings.

We have performed several additional experiments, including the quantification of multiple chloroplast buds or GFP-ATG8-labeled structures from individual plants. The results strengthen our claims and thus improve the significance of the current study. Please see the responses below for details.

Reviewer #3 (Recommendations For The Authors):

Overall, the live-cell imaging in this paper is high quality and rigorously conducted. However, without quantification of these events, it is difficult to judge whether this is an occasional contributor to plastid breakdown, or the primary mechanism for this process.

- For Figure 1, the authors could estimate the importance of this mechanism for chloroplast breakdown by calculating the volume change in chloroplasts over time during light-limiting conditions, then comparing this to the volume of the puncta that bud off of plastids and the frequency of these events. That is, what percentage of chloroplast volume loss can be accounted for by puncta that bud from chloroplasts? Are there likely other mechanisms contributing to chloroplast breakdown, or is this the primary mechanism?

We measured the volumes of chloroplast stroma when the leaves from wild-type (WT) and *atg7* plants accumulating RBCS-mRFP were subjected to extended darkness for 1 d (Figure 1-figure supplement 2). The volume of the chloroplast stroma in dark-treated leaves of WT plants was 70% that in leaves before treatment, whereas the volume of the chloroplast stroma in dark-treated *atg7* leaves was 86% that in leaves before treatment. The transport of

Rubisco-containing bodies into the vacuole did not occur in *atg7* leaves (Figure 1-figure supplement 1). These results suggest that the release of chloroplast buds as Rubisco-containing bodies contributes to the decrease in chloroplast stroma volume during dark treatment. These results also suggest that autophagy-independent systems contribute to the decrease in chloroplast volume. It is difficult to monitor the volume or frequency of budding off of puncta from chloroplasts during dark treatment because the budding and transport of the puncta occur relatively quickly and are completed within minutes, and the puncta frequently move away from the plane of focus. Additionally, continuous monitoring of chloroplast morphology over the dark treatment period requires the long-term exposure of leaves to repeated laser excitation, and such treatment might cause unexpected stress. We believe that the evaluation of chloroplast stroma volume after 1 d of dark treatment is important for estimating the contribution of the mechanism described in this study. The descriptions about this additional experiment are in lines 163–174.

- The claim that structures budding from the plastid "specifically contains stroma material...without any chlorophyll signal" (p. 6 and Figure 2) should be supported by quantitative analysis of many such buds in multiple cells from multiple independent plants.

We performed additional experiments (Figure 2-figure supplement 1) to measure the fluorescence intensity ratios of the stroma marker RBCS-GFP and chlorophyll between chloroplast budding structures and their neighboring chloroplasts in *Arabidopsis* plants expressing the stromal marker RBCS-GFP along with TOC64-mRFP (a chloroplast outer envelope membrane protein), KEA1-mRFP (a chloroplast inner envelope membrane protein), or ATPC1-tagRFP (a thylakoid membrane protein). The results indicated that chloroplast buds contain chloroplast stroma without chlorophyll signals. The descriptions of this experiment are in lines 175–199. In these experiments, we observed 30 to 33 chloroplast buds from eight individual plants.

- Claims about the dynamics of these events in Figures 2 & 3 should be supported by quantitative analysis of many buds in multiple cells from multiple independent plants and appropriate summary statistics (e.g. mean, standard deviation), and claims about the coordination of events should be supported by statistical comparison of these measurements between different markers.

As mentioned in the response to the above comments, quantification of fluorescent intensities (Figure 2-figure supplement 1) revealed that the chloroplast budding structures produced TOC64-mRFP and KEA1-mRFP signals without ATPC1-tagRFP signal. These results support the claim that chloroplast buds contain chloroplast stroma and envelope components without thylakoid membranes.

It is not easy to quantify the dynamics of chloroplast buds since the puncta sometimes move away from the plane of focus. We therefore added data from individual time-lapse observations showing that the type of movement exhibited by the puncta changes during tracking (Figure 3-figure supplement 1A and 1B, Videos 8 and 9) to strengthen the notion that such a phenomenon was observed repeatedly.

- Data in Figure 4 should be supported by quantification of the proportion of plastid-derived puncta that end up inside the vacuole (compared to those that do not) in multiple cells from multiple independent plants.

Although we performed additional observations of the destinations of chloroplast-derived puncta, we encountered some difficulty in correctly calculating the proportion of plastid-derived puncta that ended up inside the vacuole. This problem is similar to the difficulty in

tracking Rubisco-containing bodies mentioned in the response to the previous comments. During timelapse imaging, puncta sometimes move from the plane of focus toward the deeper side (abaxial side) or near side (adaxial side), causing us to lose track of a number of puncta. Therefore, we could not determine the destinations of all puncta to calculate the proportion of puncta that ended up in the vacuolar lumen.

Alternatively, we added the results of three experiments (Figure 4-figure supplement 1, Videos 12–14) examining how the vacuolar membrane engulfs the chloroplast-derived puncta to incorporate them inside the vacuole. The data support the notion that such a phenomenon occurs repeatedly in sugar-starved leaves. All results were obtained from individual plants.

- Data in Figure 6 should also be supported by quantitative analysis of many buds in multiple cells from multiple independent plants, to determine whether ATG8 associates with all RBCS-containing buds, and vice versa.

To address this issue, we performed additional experiments on plants expressing GFP-ATG8a and RBCS-mRFP (Figure 6-figure supplements 3 and 4). First, we observed 58 chloroplast buds from eight individual plants and evaluated the proportion of GFP-ATG8a-labeled chloroplast buds. We determined that 64% of chloroplast buds were at least autophagy-associated structures (Figure 6-figure supplement 3A–3C). This result also suggests that chloroplasts can form autophagy-independent budding structures, which might be associated with stromule-related structures or the autophagy-independent vesiculation machinery. We also evaluated the number of GFP-ATG8a-labeled chloroplast buds (Figure 6-figure supplement 3D and 3E). The formation of such structures increased in response to dark treatment (Figure 6-figure supplement 3D), but they did not appear in *atg7* plants exposed to the dark (Figure 6-figure supplement 3E). These results support the notion that the formation of chloroplast buds to be released as Rubisco-containing bodies requires the core ATG machinery.

Furthermore, we observed 157 GFP-ATG8a-labeled structures from thirteen individual plants and evaluated the proportion of chloroplast-associated isolation membranes (Figure 6-figure supplement 4). We also classified the chloroplast-associated, GFP-ATG8a-labeled structures into two categories: the chloroplast surface type (Figure 7-figure supplement 4A) and the chloroplast bud type (Figure 7-figure supplement 4B). This experiment suggested that 43% of the isolation membranes labeled by GFP-ATG8a were involved in chloroplast degradation during an early phase of sugar starvation (extended darkness for 5 to 9 h from the end of night) in mesophyll cells. We believe that these results indicate that autophagy contributes substantially to chloroplast degradation via the morphological changes observed in this study. The descriptions about these experiments are in lines 284–300 in the Results section and in lines 426–444 in the Discussion section.

- Which parts of the plastid bud (Fig 2), about the dynamics of the events (Fig 3), about the colocalization between ATG8 and the plastid stroma buds, and the dynamics of this association (Fig 6).

We performed multiple quantitative studies to address the issues listed above. We believe that these additional experiments strengthened our findings.

- I suggest that the authors avoid using the term "vesicles" to describe the plastid-derived puncta, since it doesn't seem like coat proteins are required for their formation. I suggest "puncta" or similar terms.

We replaced the term “vesicles” with “puncta” or other suitable terms, as suggested.

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